

Information Theory and Inference

From Shannon entropy to Hamiltonian Monte Carlo — Lecture Notes

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1. About these notes

1.1. Scope

These are lecture notes for *Information Theory and Inference* (third-semester elective, Master in Physics of Data, University of Padova, instructor Prof. Michele Allegra). The course starts from Shannon's operational definition of information and ends with modern Markov-Chain Monte Carlo methods for high-dimensional Bayesian inference.

The two halves of the course share a single organising object: the *probability distribution*. In Part I we ask how much information such a distribution contains, and how many bits are needed to communicate outcomes over a noisy channel. In Part II we ask, given a small amount of data, how to *sample* from a posterior distribution we can evaluate only up to an unknown normalisation constant.

1.2. Reading the notes

Chapters 1–5 build up Shannon's toolbox: surprisal, entropy, KL divergence, mutual information, and the two coding theorems (source and noisy-channel). Chapters 6–8 pivot to time series and computational statistics: Markov chains, MCMC, Metropolis and Gibbs sampling, with a high-dimensional inference example illustrating how the tools fit together.

Every chapter uses the styled callout environments (definition, theorem, proposition, lemma, corollary, example, remark, note, exercise, solution) so that key results can be scanned without reading the surrounding prose.

1.3. Notation

- X, Y, Z : random variables; x, y, z realisations; $\mathcal{X}$ their alphabet.
- $p_X(x) = P(X = x)$, $p(x, y)$: probability mass or density; $p(x | y)$ conditional.
- $H(X)$, $H(X, Y)$, $H(X | Y)$: entropy, joint entropy, conditional entropy.
- $I(X; Y)$: mutual information; $D(p \| q)$: Kullback–Leibler divergence.
- C : channel capacity; R : code rate.
- $\pi(\mathbf{x})$: target distribution in MCMC; $T(\mathbf{x} \rightarrow \mathbf{x}')$: transition kernel; $q(\mathbf{x}' | \mathbf{x})$: proposal.

1.4. Sources

These notes are a re-organised version of *Notes on Information Theory and Inference* by Prof. Michele Allegra (University of Padova, 2025), with additional material from the two standard textbooks:

- Thomas M. Cover & Joy A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006.
- David J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge, 2003.

Chapters 7–8 also draw on Steve Brooks, Andrew Gelman, Galin Jones & Xiao-Li Meng (eds.), *Handbook of Markov Chain Monte Carlo*, CRC Press, 2011.

1. *About these notes*

Every chapter closes with its own *Sources* section naming the specific sections used.

Part I.

Part I: Information

2. Defining information

Information theory begins with a puzzle: what does it mean, quantitatively, that one message is *more informative* than another? Shannon's answer — that the information content of an event is the logarithm of the reciprocal of its probability — turns out to have a striking number of good properties. This chapter fixes the vocabulary and derives the first quantity of interest, the entropy of a probability space.

2.1. Events and probability spaces

We assume the reader is familiar with probability. Recall the standard structure:

Definition 2.1 (Probability space). A **probability space** is a triple $(\Omega, \mathcal{F}, P)$ where Ω is the sample space, $\mathcal{F}$ a σ -algebra of events (subsets of Ω), and $P : \mathcal{F} \rightarrow [0, 1]$ a probability measure obeying the Kolmogorov axioms.

For most of what follows we work with a finite alphabet $\mathcal{X}$ and take $\Omega = \mathcal{X}$, $\mathcal{F} = 2^{\mathcal{X}}$. The atomic events are singletons $\{x\}$ with probabilities $p_X(x) = P(\{X = x\})$.

2.2. The surprisal

The right quantitative notion of “how much information does the event $\{X = x\}$ carry” turns out to be the **surprisal**.

Definition 2.2 (Surprisal). The **surprisal** (or self-information) of an event A with probability $P(A)$ is

$$I(A) = -\log_2 P(A) \quad \text{bits.}$$

For a random variable X taking value x we write $I(x) = -\log_2 p_X(x)$.

Proposition 2.1 (Characterising the surprisal). *The function $I(A) = f(P(A))$ is uniquely determined (up to the choice of logarithm base) by three properties:*

1. **Monotonicity.** *I is a decreasing function of P : rarer events are more surprising.*
2. **Non-negativity.** *$I \geq 0$, with $I = 0$ iff $P = 1$.*
3. **Additivity for independent events.** *If A, B are independent, $I(A \cap B) = I(A) + I(B)$.*

Cauchy's functional equation forces $f(p) = -\log_c p$ for some base $c > 1$; the choice $c = 2$ measures information in bits.

Remark. The base of the logarithm is a matter of units, not physics. $c = 2$ gives bits (Shannon), $c = e$ gives nats (Kolmogorov), $c = 10$ gives dits/hartleys. All results in the notes can be rewritten in any base by multiplying entropy by the appropriate constant.

2. Defining information

Example 2.1 (Coin toss)

For a fair coin, $p(H) = p(T) = 1/2$: each outcome carries $\log_2 2 = 1$ bit of information. For a biased coin with $p(H) = 0.99$, seeing heads carries only $-\log_2 0.99 \approx 0.014$ bits — the outcome was almost certain. Seeing tails carries $-\log_2 0.01 \approx 6.6$ bits — a much more informative event.

2.3. Probability trees

A useful visualisation for compound events is the **probability tree**: a rooted tree with edges labelled by conditional probabilities, whose leaves are the possible outcomes. Multiplying probabilities along a path gives the probability of the corresponding sequence.

Lemma 2.2 (Chain rule for surprisal). *For any sequence of events $X_1, X_2, \dots, X_n$,*

$$I(x_1, x_2, \dots, x_n) = \sum_{i=1}^n I(x_i \mid x_1, \dots, x_{i-1}),$$

where $I(x_i \mid \cdot) = -\log_2 p(x_i \mid \cdot)$ is the conditional surprisal.

This is a direct consequence of the chain rule $p(x_1, \dots, x_n) = \prod_i p(x_i \mid x_1, \dots, x_{i-1})$ and additivity of the logarithm.

2.4. Information for a probability space

The surprisal is the information of a *single* outcome. To characterise the information content of the whole distribution, we average over all outcomes.

Definition 2.3 (Shannon entropy). The **Shannon entropy** of a discrete random variable X with distribution p_X is

$$H(X) = -\sum_{x \in \mathcal{X}} p_X(x) \log_2 p_X(x) = \mathbb{E}[I(X)].$$

By convention $0 \log 0 = 0$.

The entropy is the *expected* number of bits of information carried by a random outcome. This is Shannon's operational reading, and it will be justified rigorously by the source-coding theorem in Chapter 3.

Example 2.2 (Entropy of Bernoulli)

For $X \sim \text{Bern}(p)$, $H(X) = -p \log_2 p - (1-p) \log_2 (1-p) \equiv h_2(p)$. The **binary entropy function** h_2 is symmetric around $p = 1/2$, where it takes its maximum value $h_2(1/2) = 1$ bit. It vanishes at $p = 0$ or $p = 1$ — no information from a certain outcome.

Example 2.3 (Uniform distribution maximises entropy)

For a variable X taking M values, the entropy is maximised when p_X is uniform, in which case $H(X) = \log_2 M$. Non-uniform distributions give strictly less entropy — a preview of the general theorem that entropy is concave in p .

2.5. Elementary properties

The entropy is well-behaved as a function of the distribution.

Proposition 2.3 (Basic properties of entropy). *For any discrete random variable X with distribution on $\mathcal{X}$:*

1. $H(X) \geq 0$, with equality iff X is deterministic.
2. $H(X) \leq \log_2 |\mathcal{X}|$, with equality iff X is uniform.
3. H is a concave function of p_X .
4. **Grouping property.** If two outcomes x_1, x_2 are merged into one with probability $p_1 + p_2$, then

$$H(X) = H(X_{\text{merged}}) + (p_1 + p_2) h_2(p_1/(p_1 + p_2)).$$

Corollary 2.4 (Entropy of a joint variable). *For a pair (X, Y) with joint distribution $p(x, y)$,*

$$H(X, Y) = - \sum_{x, y} p(x, y) \log_2 p(x, y) \leq H(X) + H(Y),$$

with equality iff X, Y are independent (this is a special case of the sub-additivity property proved in Chapter 2).

2.6. Exercises

Exercise 2.1

A fair die is rolled. Compute (i) the surprisal of the event “the outcome is 6”, (ii) the entropy of the rolling process, (iii) the surprisal of the event “the outcome is even”.

Solution

- (i) $I(\{6\}) = -\log_2(1/6) = \log_2 6 \approx 2.585$ bits.
- (ii) $H(X) = \log_2 6 \approx 2.585$ bits (uniform on 6 outcomes).
- (iii) $I(\{2, 4, 6\}) = -\log_2(3/6) = 1$ bit — exactly one bit of information (equivalent to a coin toss).

Exercise 2.2

Show that for a Bernoulli(p) variable the entropy is maximised at $p = 1/2$ by direct differentiation. What is the second derivative at the maximum?

Solution

$H'(p) = -\log_2 p + \log_2(1-p) = \log_2((1-p)/p)$. Setting $H' = 0$ gives $p = 1/2$. The second derivative is $H''(p) = -[1/(p \ln 2) + 1/((1-p) \ln 2)]$, which at $p = 1/2$ equals $-4/\ln 2 \approx -5.77$. Negative H'' confirms concavity — a global maximum.

Exercise 2.3

Verify the grouping property for a three-outcome variable with probabilities $(1/2, 1/4, 1/4)$.

2. Defining information

Solution

Direct: $H = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 1.5$ bits. Grouping the last two into a merged outcome of probability $1/2$: $H_{merged} = h_2(1/2) = 1$ bit, plus $1/2 \cdot h_2(1/2) = 0.5$ bits, total 1.5 bits.

2.7. Callouts

Note

Shannon's definition is *operational* — it derives its meaning from the coding theorems of Chapter 3. Purely axiomatic derivations (e.g. Khinchin's) are elegant but do not tell you *why* the answer is useful.

Tip

When computing entropies by hand, remember the identity $\log_2 n = \log_2 e \cdot \ln n \approx 1.4427 \ln n$: it lets you convert between nats and bits without a calculator.

Warning

The convention $0 \log 0 = 0$ is genuine, not a hack: $p \log(1/p) \rightarrow 0$ as $p \rightarrow 0^+$. But if you compute $I(x)$ for an event that has empirical probability zero from a finite sample, the value is $+\infty$, and downstream computations will be nonsense. Always regularise.

2.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 1. Cover & Thomas, Chapter 2.

3. Information-theoretic measures

Entropy is only the first in a family of information measures. This chapter introduces the three most useful ones — conditional entropy, mutual information, and KL divergence — and develops the identities and inequalities that connect them. Every result here will be used in the coding theorems of Chapters 3–4.

3.1. Properties of Shannon entropy

We first collect a few properties beyond the elementary ones of Chapter 1.

Proposition 3.1 (Concavity of entropy). *The Shannon entropy $H : \Delta_M \rightarrow [0, \log M]$ on the probability simplex is a strictly concave function of p : for any two distributions p, q and $\lambda \in [0, 1]$,*

$$H(\lambda p + (1 - \lambda)q) \geq \lambda H(p) + (1 - \lambda)H(q),$$

with equality iff $p = q$.

Lemma 3.2 (Log-sum inequality). *For non-negative reals $a_i, b_i > 0$,*

$$\sum_i a_i \log \frac{a_i}{b_i} \geq \left(\sum_i a_i \right) \log \frac{\sum_i a_i}{\sum_i b_i},$$

with equality iff a_i/b_i is constant across i .

The log-sum inequality is the workhorse behind almost every information-theoretic inequality that follows.

3.2. Kullback–Leibler divergence

The KL divergence measures how *far* one distribution p is from another q .

Definition 3.1 (Kullback–Leibler divergence). For distributions p, q on a discrete alphabet $\mathcal{X}$,

$$D(p\|q) = \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{p(x)}{q(x)}.$$

By convention $0 \log(0/q) = 0$ and $p \log(p/0) = +\infty$ for $p > 0$.

Theorem 3.3 (Gibbs’ inequality). *For any two distributions p, q on the same alphabet, $D(p\|q) \geq 0$, with equality iff $p = q$.*

Proof idea. Apply Jensen’s inequality to the convex function $-\log$ with expectation over p : $-\log \mathbb{E}_p[q/p] \leq \mathbb{E}_p[-\log(q/p)]$; the left-hand side vanishes because $\mathbb{E}_p[q(X)/p(X)] = \sum_x q(x) = 1$, so the right-hand side is non-negative.

Remark. The KL divergence is *not* a metric: it is not symmetric ($D(p\|q) \neq D(q\|p)$ in general) and does not satisfy the triangle inequality. It nonetheless functions as a “distance” in the sense that $D(p\|q) = 0 \iff p = q$, and it is convex in each argument.

Example 3.1 (*KL of Bernoullis*)

For $p \sim \text{Bern}(p_1)$ and $q \sim \text{Bern}(p_2)$,

$$D(p\|q) = p_1 \log_2 \frac{p_1}{p_2} + (1 - p_1) \log_2 \frac{1 - p_1}{1 - p_2}.$$

When $p_2 = 1/2$ this reduces to $1 - h_2(p_1)$, the “distance from uniform” of a biased coin.

3.3. Joint entropy, mutual information, conditional entropy

Given two random variables X, Y we can build several information quantities from their joint distribution.

Definition 3.2 (Joint and conditional entropy).

$$H(X, Y) = - \sum_{x, y} p(x, y) \log_2 p(x, y), \quad H(Y | X) = \sum_x p(x) H(Y | X = x).$$

The conditional entropy $H(Y | X)$ is the average uncertainty remaining in Y once X is known.

Proposition 3.4 (Chain rule for entropy).

$$H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y).$$

More generally, $H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$.

The chain rule generalises the corresponding rule for surprisal in Chapter 1.

Definition 3.3 (Mutual information). The **mutual information** between X and Y is

$$I(X; Y) = D(p(x, y) \| p(x) p(y)) = \sum_{x, y} p(x, y) \log_2 \frac{p(x, y)}{p(x)p(y)}.$$

Rewriting the KL yields three equivalent expressions:

$$I(X; Y) = H(X) + H(Y) - H(X, Y) = H(X) - H(X | Y) = H(Y) - H(Y | X).$$

Corollary 3.5 (Non-negativity of mutual information). $I(X; Y) \geq 0$, with equality iff X, Y are independent.

Example 3.2 (*Bit conveyed by a noisy channel*)

A binary symmetric channel with crossover probability q carries information $I(X; Y) = 1 - h_2(q)$ per symbol when the input is uniform. At $q = 0$ (noise-free) it carries the full 1 bit; at $q = 1/2$ (pure noise) it carries zero.

3.4. Venn-diagram of information

The three quantities $H(X)$, $H(Y)$, $H(X, Y)$ and their combinations $H(X | Y)$, $H(Y | X)$, $I(X; Y)$ obey the same set-theoretic identities as a Venn diagram of two overlapping sets. This is the pedagogical device that generations of textbooks use to remember the relations.

Remark. The Venn-diagram picture *breaks* for three or more random variables because the “interaction information” $I(X; Y; Z)$ can be negative. Don’t over-extend the analogy.

3.5. Data-processing inequality

An operational classic: post-processing cannot increase information.

Theorem 3.6 (Data-processing inequality). *If $X \rightarrow Y \rightarrow Z$ is a Markov chain (i.e. $Z \perp\!\!\!\perp X | Y$), then*

$$I(X; Y) \geq I(X; Z).$$

Equality holds iff $Y \rightarrow Z$ preserves all information about X .

Proof. Use the chain rule $I(X; Y, Z) = I(X; Y) + I(X; Z | Y) = I(X; Z) + I(X; Y | Z)$. The Markov property makes $I(X; Z | Y) = 0$, so $I(X; Y) = I(X; Z) + I(X; Y | Z) \geq I(X; Z)$.

3.6. Convexity and further identities

Proposition 3.7 (Convexity of KL). *$D(p \| q)$ is convex in the pair (p, q) : for $\lambda \in [0, 1]$,*

$$D(\lambda p_1 + (1 - \lambda)p_2 \| \lambda q_1 + (1 - \lambda)q_2) \leq \lambda D(p_1 \| q_1) + (1 - \lambda)D(p_2 \| q_2).$$

Proposition 3.8 (Concavity of mutual information). *For fixed conditional $p(y | x)$, $I(X; Y)$ is a concave function of $p(x)$. For fixed marginal $p(x)$, $I(X; Y)$ is a convex function of $p(y | x)$. This is the workhorse behind the capacity theorems.*

3.7. Exercises

Exercise 3.1

Show that $H(X, Y) \leq H(X) + H(Y)$ with equality iff X, Y are independent, using the log-sum inequality directly (not via mutual information).

Solution

Write $H(X) + H(Y) - H(X, Y) = -\sum_{x,y} p(x, y) \log[p(x)p(y)/p(x, y)]$. By the log-sum inequality applied with $a_{xy} = p(x, y)$ and $b_{xy} = p(x)p(y)$,

$$\sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \geq 1 \cdot \log \frac{1}{1} = 0.$$

So the difference is ≥ 0 with equality iff $p(x, y)/[p(x)p(y)]$ is constant — i.e. equal to 1 — iff indepen-

dence.

Exercise 3.2

Compute $I(X; Y)$ for a binary symmetric channel with error probability q and uniform input.

Solution

$H(Y) = 1$ (uniform input plus symmetric noise gives uniform output). $H(Y | X) = h_2(q)$ (conditional on the input, Y is Bernoulli(q)). So $I(X; Y) = 1 - h_2(q)$, matching the example above.

Exercise 3.3

Prove the data-processing inequality for the specific case $X \rightarrow Y \rightarrow Z$ where $Y = f(X)$ deterministically.

Solution

$H(X | Y) = 0$ because Y determines X trivially... wait, if $Y = f(X)$ then X is not determined by Y unless f is invertible. Instead: $I(X; Z) = H(X) - H(X|Z) \leq H(X) - H(X|Y, Z)$ (conditioning reduces entropy). Since the Markov property makes $H(X|Y, Z) = H(X|Y)$, we get $I(X; Z) \leq H(X) - H(X|Y) = I(X; Y)$. The DPI holds. Deterministic post-processing is a special case.

3.8. Callouts

Note

The three interpretations of mutual information — as a KL divergence, as a reduction of entropy, and as an inclusion–exclusion of joint and marginal entropies — are all useful. Which one you invoke depends on what you want to prove.

Tip

When bounding an information quantity, try (i) apply Jensen's inequality to $-\log$; (ii) apply the log-sum inequality; (iii) invoke the data-processing inequality. These three moves cover 90% of the proofs in Cover & Thomas.

Warning

The KL divergence $D(p\|q)$ is *asymmetric*. In machine learning the direction matters: $D(p\|q)$ (with p empirical) is maximum-likelihood; $D(q\|p)$ is not.

3.9. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 2. Cover & Thomas, Chapter 2. MacKay, Chapter 2.

4. Source coding: compression

Shannon's motivation for defining entropy was to answer an engineering question: given a source that emits symbols according to a distribution p , what is the minimum number of bits per symbol needed to transmit them? This chapter proves the two source-coding theorems (lossy and lossless) that make the answer $H(X)$ operational.

4.1. Two ways to compress

There is a fundamental split in the theory:

- **Lossless compression** reconstructs every original symbol exactly. Examples: gzip, PNG, FLAC.
- **Lossy compression** allows some distortion in reconstruction. Examples: JPEG, MP3, H.264.

The tools for the two problems are different, but both build on the same object: the **typical set**.

4.2. The typical set and AEP

Definition 4.1 (Typical set). Fix $\varepsilon > 0$ and $n \in \mathbb{N}$. The ε -**typical set** of length- n sequences from source $X \sim p$ is

$$A_\varepsilon^{(n)} = \left\{ x^n \in \mathcal{X}^n : \left| -\frac{1}{n} \log_2 p(x^n) - H(X) \right| \leq \varepsilon \right\}.$$

That is, sequences whose empirical entropy is close to $H(X)$.

Theorem 4.1 (Asymptotic Equipartition Property (AEP)). For $X_1, X_2, \dots$ i.i.d. $\sim p$:

1. $-\frac{1}{n} \log_2 p(X_1, \dots, X_n) \rightarrow H(X)$ in probability as $n \rightarrow \infty$.
2. $P(A_\varepsilon^{(n)}) \rightarrow 1$ as $n \rightarrow \infty$.
3. $|A_\varepsilon^{(n)}| \leq 2^{n(H(X)+\varepsilon)}$ and $|A_\varepsilon^{(n)}| \geq (1-\varepsilon)2^{n(H(X)-\varepsilon)}$ for n sufficiently large.

The AEP is the workhorse of Shannon-style theorems: almost all probability mass concentrates on a set of roughly 2^{nH} sequences, and these sequences are (asymptotically) equiprobable — hence the *equipartition*.

Remark (Physical meaning). The AEP is the information-theoretic analogue of the ergodic theorem: the “typical” behaviour of a long random sequence is the same as the ensemble-averaged behaviour, up to exponentially small corrections.

4.3. Lossy source coding

Theorem 4.2 (Shannon's source coding theorem (lossy)). Fix $\varepsilon > 0$. For n sufficiently large, there

4. Source coding: compression

exists a code $\phi_n : \mathcal{X}^n \rightarrow \{0,1\}^{n(H(X)+\varepsilon)}$ and decoder ψ_n such that

$$P(\psi_n(\phi_n(X^n)) \neq X^n) < \varepsilon.$$

Conversely, any code with rate $R < H(X)$ has error probability tending to 1.

Proof sketch. Enumerate the typical set with $\lceil n(H + \varepsilon) \rceil$ bits and send atypical sequences to a garbage symbol. By AEP the error probability tends to zero. For the converse use Fano's inequality (below) to show that any rate below H leaves a positive average distortion.

Corollary 4.3 (Rate–distortion trade-off). *Compression at rate $R < H(X)$ is possible only at the cost of a positive distortion; the rate–distortion function $R(D) = \min_{p(\hat{x}|x): \mathbb{E}d(X, \hat{X}) \leq D} I(X; \hat{X})$ characterises the optimal trade-off for distortion measure d .*

4.4. Lossless source coding: symbol codes

For truly lossless compression the constraint is that the code be uniquely decodable.

Definition 4.2 (Prefix code). A code $\phi : \mathcal{X} \rightarrow \{0,1\}^*$ is a **prefix code** if no codeword is a prefix of another. Prefix codes are instantaneously decodable: the decoder recognises each codeword as soon as its last bit arrives.

Theorem 4.4 (Kraft's inequality). *A set of codeword lengths $\ell_1, \dots, \ell_M$ can be realised by a binary prefix code iff*

$$\sum_{i=1}^M 2^{-\ell_i} \leq 1.$$

Proof. Represent codewords as leaves of a binary tree; a length- ℓ_i leaf occupies a fraction $2^{-\ell_i}$ of the tree. Prefix-free means the leaves are disjoint, so their measures sum to at most 1. Conversely, given lengths satisfying the inequality, greedily place codewords in a binary tree from shortest to longest.

Corollary 4.5 (McMillan's inequality). *Kraft's inequality holds for the strictly larger class of uniquely decodable codes — no gain in achievable code lengths by relaxing the prefix condition.*

4.5. Source coding theorem for symbol codes

Theorem 4.6 (Source coding theorem for symbol codes). *For any uniquely decodable code $\phi : \mathcal{X} \rightarrow \{0,1\}^*$ with expected length $L = \sum_x p(x)\ell_x$,*

$$L \geq H(X)/\log_2 2 = H(X) \quad \text{bits/symbol.}$$

Conversely there exists a prefix code (a Shannon–Fano code) with $L < H(X) + 1$.

Proof. Lower bound via Kraft + Gibbs' inequality; upper bound by choosing $\ell_x = \lceil \log_2(1/p(x)) \rceil$.

Example 4.1 (*Shannon–Fano codes*)

For X with distribution $(0.5, 0.25, 0.125, 0.125)$: pick $\ell = (1, 2, 3, 3)$, so codewords 0, 10, 110, 111. Expected length $L = 0.5 + 0.5 + 0.375 + 0.375 = 1.75$ bits. This exactly matches $H(X) = 1.75$ bits — a rare case where a simple construction is optimal.

4.6. Huffman codes

Shannon–Fano codes are simple but not always optimal. Huffman’s greedy algorithm produces an *optimal* prefix code for a given source distribution.

Definition 4.3 (Huffman coding algorithm). Given probabilities $p_1, \dots, p_M$:

1. Repeatedly combine the two lowest-probability symbols into a super-symbol whose probability is their sum.
2. Assign bits 0 and 1 to the merged edges.
3. Read the codeword for each leaf by traversing from root to leaf.

Theorem 4.7 (Huffman optimality). *The expected length of a Huffman code satisfies*

$$H(X) \leq L^{Huff} < H(X) + 1,$$

and no prefix code (with the same source distribution) has smaller expected length.

The gap of at most 1 bit is due to the ceiling operation. For long blocks the gap becomes negligible: coding n -tuples of source symbols achieves expected length within $1/n$ of the entropy per symbol.

Example 4.2 (*Huffman for skewed source*)

For X with probabilities $(0.4, 0.35, 0.2, 0.05)$: entropy $H = 1.74$ bits. Huffman merges $0.05 + 0.2 = 0.25$, then $0.25 + 0.35 = 0.60$, then $0.60 + 0.4 = 1.0$. Codewords: 0, 10, 110, 111 with lengths $(1, 2, 3, 3)$; expected length $L = 1.85$ bits, only 0.11 above H .

4.7. Universal codes

In practice one rarely knows the source distribution in advance. **Universal codes** (Lempel–Ziv, arithmetic coding, context-tree weighting) achieve entropy rate for the true distribution *without knowing it*: they estimate the distribution from the observed prefix and adapt on the fly.

Proposition 4.8 (LZ77 universality). *The Lempel–Ziv 1977 sliding-window algorithm, applied to a stationary ergodic source of entropy rate H_∞ , achieves an asymptotic compression rate of H_∞ per symbol as the window grows. No advance knowledge of the source is needed.*

Remark. Modern file formats (gzip, bzip2, xz) are variations on Lempel–Ziv and arithmetic coding. The theoretical guarantee that they hit the entropy limit is a Shannon-theorem consequence of the AEP for ergodic sequences.

4.8. Exercises

Exercise 4.1

Construct a Huffman code for a source with probabilities $(1/2, 1/4, 1/8, 1/16, 1/16)$ and verify it achieves $H(X)$.

Solution

$H(X) = 1 + 0.5 + 0.375 + 0.25 + 0.25 = 1.875$ bits. Huffman merges: $(1/16 + 1/16) = 1/8$, then $(1/8 + 1/8) = 1/4$, etc. Final codewords: 0, 10, 110, 1110, 1111 with lengths (1, 2, 3, 4, 4); expected length $L = 1/2 + 1/2 + 3/8 + 1/4 + 1/4 = 1.875$ bits. Exactly H — because all probabilities are dyadic.

Exercise 4.2

Show that Kraft's inequality implies $\sum_i 2^{-\ell_i} = 1$ for a *complete* prefix code (one where every internal node has two children).

Solution

Each leaf i of the binary tree at depth ℓ_i occupies a fraction $2^{-\ell_i}$ of the tree, and a complete binary tree has its leaves partitioning the entire tree, so $\sum_i 2^{-\ell_i} = 1$. An *incomplete* code (missing some leaves) gives strict inequality.

Exercise 4.3

Compute $R(D)$ for a Gaussian source with variance σ^2 and squared-error distortion $d(x, \hat{x}) = (x - \hat{x})^2$.

Solution

The rate-distortion function is $R(D) = \frac{1}{2} \log_2(\sigma^2/D)$ for $0 \leq D \leq \sigma^2$, and $R(D) = 0$ for $D \geq \sigma^2$. It is achieved by a jointly Gaussian test channel with reverse-water-filling on the eigenvalues.

4.9. Callouts

Note

The source-coding theorem is *asymptotic*: it says the entropy is achievable in the limit of long blocks. Finite-block codes always leave a gap, typically $O(\log n/n)$.

Tip

For a single-symbol Huffman code the code length is $\lceil \log_2(1/p_i) \rceil$; if this is *not* achievable, the gap is at most 1 bit and vanishes for long-block codes.

Important

Real-world compressors use *arithmetic* coding rather than Huffman when the source distribution has one very-high-probability symbol, because arithmetic coding can produce non-integer code lengths and achieve entropy exactly.

 Warning

Universal codes give the *asymptotic* compression rate. On short inputs (a single tweet) they often *expand* the data due to header overhead. Don't benchmark compressors on toy examples.

4.10. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapters 3–4. Cover & Thomas, Chapters 3, 5, 10. MacKay, Chapters 4, 5, 6.

5. Noisy channels and coding

Shannon's second big theorem answers a complementary question: given a noisy channel, what is the maximum rate at which information can be transmitted with vanishing error probability? The answer is the channel *capacity* $C = \max_{p(x)} I(X; Y)$, and its achievability is proved by the beautiful *random-coding* argument.

5.1. Decoding as inference

Consider a channel $p(y | x)$: an input symbol x is transmitted and a possibly-corrupted output y is observed. The decoder's task is to *infer* what was sent given what was received.

Definition 5.1 (Discrete memoryless channel). A **discrete memoryless channel (DMC)** is a triple $(\mathcal{X}, \mathcal{Y}, p(y | x))$: input alphabet $\mathcal{X}$, output alphabet $\mathcal{Y}$, and a conditional distribution $p(y | x)$. Memoryless means for a length- n transmission,

$$p(y^n | x^n) = \prod_{i=1}^n p(y_i | x_i).$$

Example 5.1 (Binary symmetric channel)

The BSC with crossover probability $q \in [0, 1/2]$ has $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ and $p(1 | 0) = p(0 | 1) = q$. It flips each bit independently with probability q .

Example 5.2 (Binary erasure channel)

The BEC with erasure probability ε has $\mathcal{Y} = \{0, 1, ?\}$ and $p(? | x) = \varepsilon$, $p(x | x) = 1 - \varepsilon$. Each input is either received correctly or lost — but *never* silently corrupted.

5.2. Error-correcting codes

The idea of coding is to send a *codeword* $\mathbf{x} = (x_1, \dots, x_n)$ chosen from a codebook $\mathcal{C} \subset \mathcal{X}^n$ of size $|\mathcal{C}| = 2^{nR}$, so the code rate is $R = (\log_2 |\mathcal{C}|)/n$ bits per channel use. On the receiver side, a decoder $g : \mathcal{Y}^n \rightarrow \mathcal{C}$ maps the received sequence back to a codeword.

Definition 5.2 (Achievable rate). A rate R is **achievable** on channel $p(y | x)$ if there exists a sequence of codes $\{\mathcal{C}_n\}$ of size $|\mathcal{C}_n| = 2^{nR}$ and decoders $\{g_n\}$ such that the maximal error probability

$$\lambda^{(n)} = \max_{\mathbf{x} \in \mathcal{C}_n} P(g_n(Y^n) \neq \mathbf{x} | X^n = \mathbf{x})$$

tends to zero as $n \rightarrow \infty$. The **capacity** C is the supremum of achievable rates.

5.3. The noisy-channel coding theorem

Theorem 5.1 (Shannon's noisy-channel coding theorem). *For a discrete memoryless channel,*

$$C = \max_{p(x)} I(X; Y).$$

Any rate $R < C$ is achievable, and any rate $R > C$ is not.

Proof idea (achievability). Fix an input distribution $p(x)$ achieving $I(X; Y)$. Generate a random codebook of 2^{nR} i.i.d. $\sim p^{\otimes n}$ codewords. Decode by *joint typicality*: given Y^n , output the unique codeword $\hat{X}^n$ such that $(X^n, Y^n) \in A_\varepsilon^{(n)}$. The average error probability, averaged over both the message and the random codebook, tends to zero for $R < C - 2\varepsilon$. Then expurgate half the codebook to get a code with *maximal* error probability tending to zero.

Proof idea (converse). Use Fano's inequality: $H(\mathbf{X} | \hat{\mathbf{X}}) \leq 1 + P_e \cdot nR$, which combined with data processing gives $nR \leq nC + 1 + nRP_e$, so $R \leq C$ if $P_e \rightarrow 0$.

Example 5.3 (BSC and BEC capacities)

For the BSC with crossover q , the capacity is $C = 1 - h_2(q)$, achieved by uniform $p(x)$. For the BEC with erasure ε , $C = 1 - \varepsilon$: each non-erased symbol conveys one bit.

5.4. Fano's inequality

Theorem 5.2 (Fano's inequality). *Let X take values in $\mathcal{X}$ and let $\hat{X}$ be an estimate of X from any random variable Y . Let $P_e = P(\hat{X} \neq X)$. Then*

$$H(X | Y) \leq H(X | \hat{X}) \leq h_2(P_e) + P_e \log_2(|\mathcal{X}| - 1).$$

Fano's inequality lower-bounds the error probability of any estimator: if P_e is small then $H(X | Y)$ must be small too. It is the fundamental tool for converse coding theorems throughout the notes.

Corollary 5.3 (Rate-distortion converse via Fano). *Combining Fano with the data-processing inequality yields the converse of the noisy-channel coding theorem: any code with rate above capacity has average error probability bounded away from zero.*

5.5. Random coding

The genius of Shannon's proof is to show existence by averaging. The typical codebook is bad in ways that are hard to see, but the *ensemble* of codebooks has good average behaviour — so at least *one* codebook must be at least as good as the average.

Proposition 5.4 (Random-coding exponent). *For $R < C$, the average error probability over random codebooks decays exponentially in n :*

$$\bar{P}_e \leq 2^{-n(E_r(R))},$$

where $E_r(R) > 0$ for $R < C$ is Gallager's random-coding exponent. This gives the sharp asymptotic rate of convergence, not just the fact that it holds.

Remark. The random-coding exponent is what makes Shannon’s theorem *quantitative*: it tells you how large n has to be for a given target error. Modern coding theory is largely the search for structured codes that meet or approach the random-coding exponent with efficient encoders and decoders (turbo codes, LDPC, polar codes).

5.6. Typical-set decoding

Definition 5.3 (Jointly typical sequences). Given $p(x, y)$, the jointly typical set $A_\varepsilon^{(n)}(X, Y)$ is

$$\left\{ (x^n, y^n) : \left| -\frac{1}{n} \log p(x^n) - H(X) \right| < \varepsilon, \left| -\frac{1}{n} \log p(y^n) - H(Y) \right| < \varepsilon, \left| -\frac{1}{n} \log p(x^n, y^n) - H(X, Y) \right| < \varepsilon \right\}.$$

Lemma 5.5 (Joint AEP). As $n \rightarrow \infty$: (i) $P((X^n, Y^n) \in A_\varepsilon^{(n)}) \rightarrow 1$; (ii) $|A_\varepsilon^{(n)}(X, Y)| \leq 2^{n(H(X, Y) + \varepsilon)}$; (iii) if $(\tilde{X}^n, \tilde{Y}^n)$ are drawn independently from the marginals, then $P((\tilde{X}^n, \tilde{Y}^n) \in A_\varepsilon^{(n)}) \leq 2^{-n(I(X; Y) - 3\varepsilon)}$.

Property (iii) is the *key* to the noisy-channel coding theorem: it says that if you draw two random n -sequences from the marginals, they almost never look typical together — the probability of a “spurious match” is exponentially small in $nI(X; Y)$.

5.7. Structured codes

The random-coding proof does not construct a specific code; it shows one exists. Practical coding requires structured codes with fast decoders.

Example 5.4 (Repetition code)

Send each bit n times, majority-decode. For a BSC(q) this has rate $R = 1/n$ and error probability $\sim \exp(-nD(1/2\|q))$. Poor rate but easy analysis; a textbook baseline.

Example 5.5 (Hamming (7, 4) code)

Four data bits, three parity bits, minimum distance 3: corrects any single error in a block of 7. Rate $R = 4/7$; a first example of an *algebraic* code that meets the Singleton bound.

Modern codes — LDPC (Gallager 1962), turbo codes (Berrou et al. 1993), polar codes (Arikan 2009) — approach capacity within fractions of a dB and are decoded in polynomial time. Their invention represents 60 years of work catching up with Shannon.

5.8. Exercises

Exercise 5.1

Compute the capacity of the Z-channel: $\mathcal{X} = \mathcal{Y} = \{0, 1\}$, $p(0 | 0) = 1$, $p(0 | 1) = q$, $p(1 | 1) = 1 - q$.

Solution

Let the input distribution be $p(0) = \alpha$, $p(1) = 1 - \alpha$. Then $H(Y | X) = (1 - \alpha)h_2(q)$ and $H(Y) = h_2(\alpha + (1 - \alpha)q)$ (Bernoulli output). Maximise $I = H(Y) - H(Y | X)$ over α . The optimum satisfies a

transcendental equation solvable numerically; e.g. for $q = 0.5$ the capacity is ≈ 0.322 bit per channel use, less than the BSC(0.5) capacity (which is zero) — the Z-channel is asymmetric and therefore useful even at large error rates.

Exercise 5.2

Prove Fano's inequality using the identity $H(X, \hat{X}) = H(\hat{X}) + H(X | \hat{X})$ and a careful decomposition of $H(X | \hat{X})$ into “the error event” and “given the error event”.

Solution

Introduce indicator $E = \mathbf{1}[\hat{X} \neq X]$. Then $H(E, X | \hat{X}) = H(E | \hat{X}) + H(X | E, \hat{X}) \leq h_2(P_e) + P_e \log_2(|\mathcal{X}| - 1)$ (given error, X is one of the $|\mathcal{X}| - 1$ remaining values). Since $H(X | \hat{X}) \leq H(E, X | \hat{X})$, the inequality follows.

5.9. Callouts

Note

The noisy-channel coding theorem is *the* result of information theory: it establishes an operational meaning of mutual information as the maximum rate of reliable communication.

Tip

For an asymmetric channel it is often useful to first find $p(x)$ that maximises $I(X; Y)$ numerically, then compute the resulting capacity. Symmetric channels admit closed-form solutions.

Warning

Shannon's theorem is an existence result. Constructing capacity-achieving codes was an open problem for decades; even now the practical gap is a few tenths of a dB for high-noise regimes.

Important

The capacity C is an asymptotic (long-block) quantity. Finite-block coding has a *lower* achievable rate, characterised by the finite-block-length coding bounds of Polyanskiy, Poor & Verdú (2010).

5.10. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 5. Cover & Thomas, Chapters 7, 8. MacKay, Chapters 9–11. Modern converse: Polyanskiy, Poor & Verdú, *Channel coding rate in the finite blocklength regime*, IEEE Trans. Inf. Theory **56**, 2307 (2010).

6. Continuous random variables

Shannon's theory extends naturally from discrete to continuous alphabets, but with a twist: continuous entropies can be negative, and the operational meaning of *differential* entropy is more subtle than that of Shannon entropy. This chapter introduces the differential entropy, the mutual information of continuous variables, and the Gaussian channel — the paradigm of a continuous-alphabet system.

6.1. Differential entropy

Definition 6.1 (Differential entropy). For a random variable X with density $f_X(x)$, the **differential entropy** is

$$h(X) = - \int f_X(x) \log_2 f_X(x) dx,$$

provided the integral exists.

The formula is the natural continuous analogue of the discrete Shannon entropy, but its interpretation is different:

Remark. $h(X)$ is *not* the average number of bits to describe X : continuous variables need infinitely many bits to describe with arbitrary precision. Instead, $h(X)$ measures the entropy of X *relative to a uniform reference on unit intervals*: quantising X to precision Δ gives a discrete variable with entropy $h(X) - \log_2 \Delta$. The negative differential entropy of a very concentrated density corresponds to a discrete variable that is easier to encode than a uniform bin of size Δ .

Example 6.1 (Uniform and Gaussian differential entropies)

For $X \sim \text{Unif}[0, a]$: $h(X) = \log_2 a$. Note $h < 0$ if $a < 1$.

For $X \sim \mathcal{N}(\mu, \sigma^2)$: $h(X) = \frac{1}{2} \log_2(2\pi e \sigma^2)$, independent of μ .

6.2. Basic properties

Many properties of Shannon entropy carry over unchanged, but with two important exceptions:

Proposition 6.1 (Properties of differential entropy). 1. $h(X)$ can be negative (see uniform example).
2. *Scaling*: $h(aX) = h(X) + \log_2 |a|$; *translation*: $h(X + c) = h(X)$.
3. *Sub-additivity*: $h(X, Y) \leq h(X) + h(Y)$, equality iff X, Y independent.
4. *Chain rule*: $h(X, Y) = h(X) + h(Y | X)$.
5. *Conditioning reduces entropy*: $h(Y | X) \leq h(Y)$.

6.3. Maximum-entropy distributions

Discrete entropies are maximised by the uniform distribution. Continuous ones are maximised by distributions determined by the given constraints:

Theorem 6.2 (Gaussian maximises entropy under variance constraint). *Among all real random variables with variance σ^2 , the Gaussian $\mathcal{N}(\mu, \sigma^2)$ has the largest differential entropy $h = \frac{1}{2} \log_2(2\pi e \sigma^2)$.*

Proof. Use Gibbs' inequality: for any density f with the given variance,

$$D(f \| f_{\mathcal{N}}) = h(f_{\mathcal{N}}) - h(f) \geq 0,$$

where the identity uses only $\int f x^2 dx = \sigma^2$ and $\int f = 1$.

Corollary 6.3 (Other max-ent distributions). • *Bounded support $[a, b]$: uniform maximises h with $h = \log_2(b - a)$.*

- *Non-negative support, fixed mean μ : exponential maximises h with $h = \log_2(e \mu)$.*
- *Fixed first and second moments: Gaussian maximises.*

6.4. Mutual information for continuous variables

Definition 6.2 (Continuous mutual information). For jointly-distributed (X, Y) with joint density $f(x, y)$,

$$I(X; Y) = \iint f(x, y) \log_2 \frac{f(x, y)}{f(x) f(y)} dx dy.$$

Unlike differential entropy, $I(X; Y) \geq 0$ for continuous variables as well — the KL-divergence structure survives the limit.

Proposition 6.4 (Mutual information under transformation). *$I(X; Y)$ is invariant under invertible smooth transformations of X and Y separately: for diffeomorphisms ϕ, ψ ,*

$$I(\phi(X); \psi(Y)) = I(X; Y).$$

This is what makes I a reparameterisation-invariant measure of dependence.

6.5. The Gaussian channel

Definition 6.3 (Gaussian channel). The **additive Gaussian noise channel** with power constraint P is

$$Y = X + Z, \quad Z \sim \mathcal{N}(0, N), \quad \mathbb{E}[X^2] \leq P.$$

The average power of the transmitted signal is bounded by P ; the noise has variance N ; both are per channel use.

Theorem 6.5 (Gaussian channel capacity). *The capacity of the Gaussian channel is*

$$C = \frac{1}{2} \log_2(1 + P/N) \quad \text{bits/channel use.}$$

The maximising input distribution is $X \sim \mathcal{N}(0, P)$.

Proof. Bound $I(X; Y) = h(Y) - h(Y | X) = h(Y) - h(Z)$. Since $\text{Var}(Y) = \text{Var}(X) + N \leq P + N$ and Gaussian maximises entropy for a given variance, $h(Y) \leq \frac{1}{2} \log_2(2\pi e(P + N))$, with equality iff Y Gaussian, iff X Gaussian. Then $C = \frac{1}{2} \log_2(2\pi e(P + N)) - \frac{1}{2} \log_2(2\pi e N) = \frac{1}{2} \log_2(1 + P/N)$.

Example 6.2 (*Bandwidth-limited channel*)

For a channel of bandwidth B Hz and noise spectral density $N_0/2$, the (Shannon–Hartley) capacity is

$$C = B \log_2(1 + P/(N_0 B)) \quad \text{bits/second.}$$

This is the celebrated formula relating bandwidth, SNR, and channel capacity that governs everything from modem design to DSL to Wi-Fi.

6.6. Parallel Gaussian channels and water-filling

For a bank of K parallel independent Gaussian channels with individual noise variances N_i and a total power budget $\sum P_i \leq P$:

Theorem 6.6 (Water-filling). *The capacity is*

$$C = \sum_{i=1}^K \frac{1}{2} \log_2(1 + P_i/N_i),$$

maximised by allocating power $P_i = \max(0, \mu - N_i)$, where μ is the water level chosen so that $\sum_i P_i = P$.

The intuition: pour a total volume P of “water” over containers of depths N_i ; deeper containers (noisier channels) receive less water.

Remark (Water-filling in modern communications). Water-filling is what OFDM, DSL, LTE, and 5G do at the physical layer: split the frequency band into sub-carriers with individual SNRs, allocate transmit power to maximise total throughput.

6.7. Exercises

Exercise 6.1

Compute the differential entropy of a Laplace(μ, b) distribution $f(x) = (2b)^{-1} \exp(-|x - \mu|/b)$.

Solution

$h(X) = -\int f \log_2 f = -\int f \log_2 \frac{1}{2b} + \log_2 e \int f |x - \mu|/b dx = \log_2(2b) + \log_2 e$. So $h = \log_2(2be)$ bits. Interesting comparison: for the same variance $2b^2$, the Gaussian has entropy $\frac{1}{2} \log_2(2\pi e \cdot 2b^2) = \frac{1}{2} \log_2(4\pi eb^2)$; Laplace has $h = \log_2(2be) = \frac{1}{2} \log_2(4b^2 e^2)$. Ratio: Gaussian is larger by factor $\log_2(\pi/e) \approx 0.20$ bits — Gaussian is the max-ent distribution.

Exercise 6.2

For the Gaussian channel with SNR = 10 dB, compute the capacity in bits/s/Hz.

Solution

$C = \log_2(1 + 10) = \log_2 11 \approx 3.46$ bits/s/Hz. Modern LTE aims to reach $\sim 90\%$ of this in the physical layer with efficient turbo/LDPC coding.

Exercise 6.3

Verify that among Gaussian variables the Fisher information $J(f) = \mathbb{E}[(\partial_x \log f)^2]$ satisfies $J(f) = 1/\sigma^2$ and that the entropy-Fisher information identity $J(f) \geq 2\pi e \cdot 2^{-2h(f)}$ holds with equality iff Gaussian.

Solution

Direct: $\partial_x \log \mathcal{N}(x; 0, \sigma^2) = -x/\sigma^2$, so $J = 1/\sigma^2$. Substituting the Gaussian $h = \frac{1}{2} \log_2(2\pi e \sigma^2)$: $2\pi e \cdot 2^{-2h} = 2\pi e / (2\pi e \sigma^2) = 1/\sigma^2 = J$. This is the Stam / Blachman inequality — a Cramér–Rao-like bound in disguise.

6.8. Callouts**i Note**

Differential entropy is a *coordinate-dependent* quantity: change of variable $Y = g(X)$ gives $h(Y) = h(X) + \mathbb{E}[\log |g'(X)|]$. Only quantities like $I(X; Y)$ and KL are coordinate-free.

💡 Tip

When you need entropy for a continuous variable, first check whether it fits a maximum-entropy family (Gaussian, exponential, uniform); those admit closed-form answers. Otherwise, quantise and use the discrete entropy formula.

⚠ Warning

Negative differential entropy is *not* a sign of an error: it just means the distribution is more concentrated than a unit-support uniform. It becomes a positive quantity again when converted to a discrete entropy at any finite resolution.

! Important

The Gaussian capacity $\frac{1}{2} \log_2(1 + \text{SNR})$ is *the* practical benchmark of digital communication systems. Anything above it violates Shannon; anything close to it is state-of-the-art coding.

6.9. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 6. Cover & Thomas, Chapters 8–9. MacKay, Chapter 11.

Part II.

Part II: Inference

7. Markov chains and information in time series

The information framework of the previous chapters treated symbols as i.i.d. Real data are almost never i.i.d.: language has grammar, DNA has motifs, prices have trends. To measure the information content of a sequence with memory we need to consider *stochastic processes* — Markov chains are the simplest interesting case.

7.1. Markov chains

Definition 7.1 (Markov chain). A discrete-time **Markov chain** on state space $\mathcal{S}$ is a sequence $\{X_t\}_{t \geq 0}$ such that

$$P(X_{t+1} = s' \mid X_t = s, X_{t-1}, \dots, X_0) = P(X_{t+1} = s' \mid X_t = s) = T(s, s'),$$

where T is the **transition matrix**. The distribution at time t evolves as $\pi_{t+1} = \pi_t T$.

Definition 7.2 (Stationary distribution). A distribution π^* is **stationary** for T if $\pi^* = \pi^* T$. If additionally

$$\pi^*(s) T(s, s') = \pi^*(s') T(s', s) \quad \forall s, s',$$

the chain satisfies **detailed balance** and is called *reversible*.

Theorem 7.1 (Convergence to stationarity). *An irreducible, aperiodic Markov chain on a finite state space has a unique stationary distribution π^* , and $\pi_t \rightarrow \pi^*$ geometrically as $t \rightarrow \infty$, from any initial distribution.*

7.2. Entropy rate

For a stochastic process the entropy of a single symbol undersells the information: correlations between successive symbols mean each new symbol conveys less information than the marginal entropy suggests.

Definition 7.3 (Entropy rate). The **entropy rate** of a stationary process $\{X_t\}$ is

$$H_\infty(X) = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, \dots, X_n) = \lim_{n \rightarrow \infty} H(X_n \mid X_1, \dots, X_{n-1}),$$

whenever the limit exists. For a Markov chain the entropy rate simplifies to

$$H_\infty(X) = H(X_2 \mid X_1) = \sum_s \pi^*(s) H(T(s, \cdot)).$$

Example 7.1 (Entropy rate of English text)

Shannon's own experiments (1951) estimated the entropy rate of English text at $\sim 1\text{--}1.5$ bits per letter — much lower than the naive $\log_2 26 \approx 4.7$ bits/letter — because of strong statistical depen-

dencies between letters, words and grammar. This is exactly the compression ratio ($\sim 5\text{--}6\times$) that modern text compressors achieve.

Remark. The entropy rate is the *asymptotic* compression bound: a universal code can compress a long ergodic sequence at rate arbitrarily close to H_∞ , but no less.

7.3. Entropy production in Markov chains

Physicists care about a related quantity: the entropy *produced* by a non-equilibrium chain per step.

Definition 7.4 (Entropy production). For a Markov chain with transition matrix T and stationary distribution π^* , the **entropy production rate** is

$$\sigma = \sum_{s,s'} \pi^*(s) T(s, s') \log \frac{T(s, s')}{T(s', s)}.$$

Equivalently, $\sigma = \sum_{s,s'} \pi^*(s) T(s, s') \log \frac{\pi^*(s) T(s, s')}{\pi^*(s') T(s', s)}$: the KL divergence between forward and reverse trajectory distributions.

Theorem 7.2 (Entropy production non-negative). $\sigma \geq 0$, with equality iff detailed balance holds. Non-zero entropy production is the information-theoretic signature of non-equilibrium (breaking of time-reversal symmetry).

7.4. Hidden Markov models

Often the state of interest is not directly observable: a hidden state S_t emits an observed symbol Y_t through an emission distribution $p(y | s)$. This is a **hidden Markov model** (HMM).

Definition 7.5 (Hidden Markov model). An **HMM** is a pair $(\{S_t\}, \{Y_t\})$ where $\{S_t\}$ is a Markov chain on $\mathcal{S}$ and each Y_t is emitted independently given S_t from a distribution $p(y | s)$.

Standard HMM problems — inference of S_t from $Y_1, \dots, Y_T$ (forward-backward, Viterbi) and learning T and emission parameters (Baum–Welch, i.e. EM) — are all instances of general inference algorithms on trees or chains, and their information-theoretic complexity is well-studied.

Example 7.2 (HMM for DNA sequences)

Gene-finding algorithms model the DNA as an HMM with hidden states $S_t \in \{\text{exon, intron, intergenic, ...}\}$ and observed nucleotides $Y_t \in \{A, C, G, T\}$. Viterbi decoding assigns the most likely state sequence — a genomic annotation.

7.5. Spectrum of a Markov chain

The rate at which a Markov chain approaches stationarity is set by the second-largest eigenvalue of its transition matrix:

Proposition 7.3 (Spectral gap and mixing). *For a reversible chain with eigenvalues $1 = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_{|\mathcal{S}|}$, the total-variation distance to stationarity satisfies*

$$\|\pi_t - \pi^*\|_{TV} \leq \frac{1}{2} \sqrt{|\mathcal{S}|} \lambda_2^t.$$

*The **spectral gap** $1 - \lambda_2$ controls the rate of convergence; the **mixing time** is $t_{mix} \sim (\log |\mathcal{S}|)/(1 - \lambda_2)$.*

7.6. Convergence and independence

For an ergodic chain successive states become approximately independent at long lags, in the sense that

$$I(X_t; X_{t+k}) \rightarrow 0 \text{ as } k \rightarrow \infty,$$

at a rate governed by the spectral gap. This *mixing* property is what makes long-run averages over a chain trajectory approximate expectations under π^* — the basis of Markov Chain Monte Carlo (Chapter 7).

Corollary 7.4 (Ergodic theorem for Markov chains). *For an ergodic chain with stationary distribution π^* and any bounded $f : \mathcal{S} \rightarrow \mathbb{R}$,*

$$\frac{1}{n} \sum_{t=1}^n f(X_t) \rightarrow \sum_s \pi^*(s) f(s) = \mathbb{E}_{\pi^*}[f]$$

almost surely as $n \rightarrow \infty$.

7.7. Autocorrelation

The autocorrelation of a Markov-chain observable decays exponentially in the lag:

Proposition 7.5 (Autocorrelation of Markov chain). *For a reversible chain, the autocorrelation of f satisfies*

$$C_f(k) = \mathbb{E}_{\pi^*}[f(X_0)f(X_k)] - \mathbb{E}_{\pi^*}[f]^2 = \sum_{i \geq 2} \langle f, e_i \rangle^2 \lambda_i^k,$$

where $\{e_i\}$ are the eigenvectors of T orthonormalised in $\ell^2(\pi^)$. The dominant term decays as λ_2^k .*

Remark (Effective sample size). The **integrated autocorrelation time** $\tau_f = 1 + 2 \sum_{k \geq 1} C_f(k)/C_f(0)$ measures how many “independent” samples an n -step chain provides. The *effective sample size* is n/τ_f . Diagnosing large τ_f is the main tool for identifying poor MCMC mixing.

7.8. Exercises

Exercise 7.1

For a two-state Markov chain on $\{0, 1\}$ with transition probabilities $T(0, 1) = a$ and $T(1, 0) = b$, find the stationary distribution and entropy rate.

Solution

$\pi^*(0) = b/(a+b)$, $\pi^*(1) = a/(a+b)$. Entropy rate:

$$H_\infty = \frac{b}{a+b} h_2(a) + \frac{a}{a+b} h_2(b).$$

Exercise 7.2

Compute the entropy production of a three-state cyclic chain with rates $T(0 \rightarrow 1) = T(1 \rightarrow 2) = T(2 \rightarrow 0) = p$ (forward) and equal q (backward), $p+q < 1$.

Solution

By symmetry $\pi^* = (1/3, 1/3, 1/3)$, and

$$\sigma = 3 \cdot \frac{1}{3} \cdot p \log \frac{p}{q} + 3 \cdot \frac{1}{3} \cdot q \log \frac{q}{p} = (p-q) \log(p/q).$$

Note $\sigma > 0$ iff $p \neq q$, and equals zero iff the chain is at detailed balance.

Exercise 7.3

Show that for a reversible chain, λ_2 satisfies $\lambda_2 = 1 - 2\lambda_2^{eff}$ where λ_2^{eff} is the smallest positive eigenvalue of the graph Laplacian of T . Interpret physically.

Solution

Detailed balance $\pi^*(s)T(s, s') = \pi^*(s')T(s', s)$ makes the symmetric operator $L = I - T$ self-adjoint in $\ell^2(\pi^*)$; its spectrum is $\{1 - \lambda_i\}$. The gap $1 - \lambda_2$ is the smallest positive eigenvalue of the graph Laplacian — a Cheeger-type quantity that lower-bounds the conductance of the chain. Physically it is the *relaxation rate* of the slowest mode.

7.9. Callouts

i Note

Markov chains give an *analytical* framework for time series that fits neatly into information theory. Real time series (financial, biological) are only approximately Markov, but the framework is a useful null model.

💡 Tip

When estimating an entropy rate from data, always use a *conditional* entropy estimator: naive plug-in estimation of $H(X_1, \dots, X_n)$ has severe bias in high dimensions.

⚠ Warning

Long correlations kill mixing: chains with λ_2 very close to 1 need exponentially many samples to converge and are effectively unusable as samplers. Always diagnose with τ_f before trusting an MCMC estimate.

7.10. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 7. Cover & Thomas, Chapter 4. MacKay, Chapter 29 (for MCMC preview).

8. Markov Chain Monte Carlo and Metropolis

Suppose we want to compute an expectation $\mathbb{E}_\pi[f]$ against a distribution π we can evaluate only up to a normalisation constant — a Bayesian posterior, or a Boltzmann distribution in high dimension. Direct integration is impossible; independent sampling is impossible; but *dependent* sampling via a cleverly-designed Markov chain works. This chapter introduces the Metropolis algorithm and its theoretical justification.

8.1. The Monte Carlo approach

The core Monte Carlo idea is dead simple:

Definition 8.1 (Monte Carlo estimator). Given i.i.d. samples $X_1, \dots, X_n \sim \pi$ and a function $f : \mathcal{X} \rightarrow \mathbb{R}$ with finite variance, the **Monte Carlo estimator** of $\mathbb{E}_\pi[f]$ is

$$\hat{\mathbb{E}}_n[f] = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

By the law of large numbers $\hat{\mathbb{E}}_n[f] \rightarrow \mathbb{E}_\pi[f]$ a.s.; by the CLT the error is $O(1/\sqrt{n})$.

The rate $1/\sqrt{n}$ is *dimension-free* — it does not degrade with the size of $\mathcal{X}$. This is the fundamental advantage of Monte Carlo over deterministic quadrature in high dimension.

8.2. Simple ways to sample a distribution

For low-dimensional or well-structured π there are direct sampling algorithms:

Example 8.1 (*Inverse-CDF sampling*)

For a real random variable X with CDF F : draw $U \sim \text{Unif}(0, 1)$ and set $X = F^{-1}(U)$. Then $X \sim F$. Requires F^{-1} in closed form or via numerical inversion.

Example 8.2 (*Rejection sampling*)

To sample π , choose a proposal q with $\pi(x) \leq M q(x)$ for some $M \geq 1$. Draw $X \sim q$ and accept with probability $\pi(X)/(Mq(X))$. Accepted samples are exactly $\sim \pi$. Acceptance probability is $1/M$; the method fails in high dimension because M grows exponentially.

Remark. Direct methods fail on the problem that motivates most of Bayesian statistics: sampling from a high-dimensional posterior $\pi(\theta \mid \mathbf{y}) \propto p(\mathbf{y} \mid \theta)p(\theta)$ where the normalisation is intractable. This is where MCMC comes in.

8.3. Markov Chain Monte Carlo

The MCMC insight: construct a Markov chain whose stationary distribution is π , run it long enough that its state distribution has converged to π , and use its trajectory to estimate expectations. The samples $X_1, X_2, \dots$ are *not* independent, but by the ergodic theorem (Chapter 6) time averages still converge to expectations.

Definition 8.2 (MCMC estimator). Given a Markov chain with stationary distribution π and initial burn-in b ,

$$\hat{\mathbb{E}}_n[f] = \frac{1}{n-b} \sum_{t=b+1}^n f(X_t) \rightarrow \mathbb{E}_\pi[f] \text{ a.s.}$$

The finite-sample error scales as $\sqrt{\tau_f}/\sqrt{n}$ where τ_f is the integrated autocorrelation time of f (Chapter 6).

8.4. The Metropolis algorithm

The workhorse MCMC algorithm is Metropolis (1953) and its Hastings extension:

Definition 8.3 (Metropolis–Hastings). Given a target π known up to a constant and a proposal $q(y | x)$:

1. From current state $X_t = x$, draw a proposal $y \sim q(\cdot | x)$.
2. Compute the acceptance ratio

$$A(x \rightarrow y) = \min \left(1, \frac{\pi(y) q(x | y)}{\pi(x) q(y | x)} \right).$$

3. Set $X_{t+1} = y$ with probability $A(x \rightarrow y)$; otherwise $X_{t+1} = x$.

For a symmetric proposal ($q(y | x) = q(x | y)$), the ratio reduces to $A = \min(1, \pi(y)/\pi(x))$.

Theorem 8.1 (Metropolis correctness). *The Metropolis–Hastings kernel $T(x, y) = q(y | x)A(x \rightarrow y)$ satisfies detailed balance with respect to π : $\pi(x)T(x, y) = \pi(y)T(y, x)$. Hence π is stationary for the chain.*

Proof. Direct computation from the definition of A : the min preserves detailed balance because the acceptance ratio is exactly the ratio of forward to backward flux under $\pi \otimes q$.

Example 8.3 (Metropolis for a 1D Gaussian)

Target $\pi(x) \propto e^{-x^2/2}$. Proposal: $y = x + \sigma_p \xi$ with $\xi \sim \mathcal{N}(0, 1)$. Acceptance: $A = \min(1, e^{(x^2 - y^2)/2})$. For $\sigma_p \approx 2.4$ the chain has $\sim 24\%$ acceptance and mixes well — this is the theoretical optimum for a 1D Gaussian target.

8.5. Tuning the proposal

The step size σ_p of a Gaussian random-walk proposal has a sweet spot:

Proposition 8.2 (Optimal scaling). *For a Gaussian random-walk Metropolis on a d -dimensional target with roughly Gaussian shape, the asymptotically-optimal acceptance rate is $\alpha^* \approx 0.234$, achieved by scaling $\sigma_p^2 \propto d^{-1}$.*

Remark. The rule of thumb “aim for 20–40% acceptance” comes from this result and works surprisingly well across problems. Too-small σ_p gives $\alpha \rightarrow 1$ but the chain crawls; too-large gives $\alpha \rightarrow 0$ and the chain sticks at the current state.

8.6. Practical Metropolis

Example 8.4 (Metropolis for a Bayesian regression)

Suppose we want to sample the posterior of a Bayesian linear regression $y_i = \beta^\top \mathbf{x}_i + \varepsilon_i$ with a Gaussian likelihood and an informative Gaussian prior on β . The unnormalised posterior is

$$\pi(\beta) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_i (y_i - \beta^\top \mathbf{x}_i)^2 - \frac{1}{2\tau^2} \|\beta\|^2\right).$$

A Gaussian random-walk Metropolis proposal on β with $\sigma_p^2 = 2.4^2/d$ times a preconditioner will typically converge in $\sim 10^4$ iterations.

8.7. Multi-move / block updates

For high-dimensional targets, updating all coordinates jointly in a random-walk Metropolis has poor scaling. Two remedies:

Definition 8.4 (Component-wise Metropolis). Update each coordinate x_j separately, holding the others fixed. Each move accepts or rejects based on the conditional target $\pi(x_j \mid x_{-j})$.

Component-wise updates are the *Gibbs sampler* of Chapter 8 in the limit that the conditional proposal is the conditional target.

Proposition 8.3 (Hamiltonian Monte Carlo (HMC)). *For a target of the form $\pi(\mathbf{x}) \propto e^{-U(\mathbf{x})}$, HMC augments the state with a momentum variable and proposes $\mathbf{x} \rightarrow \mathbf{x}'$ by running Hamilton’s equations of the augmented system, then does a Metropolis accept/reject on the joint energy. The proposal preserves the target and mixes at a rate that scales as $O(d^{1/4})$ rather than $O(d^{-1})$ for random walk.*

Remark. HMC / NUTS is the algorithm behind Stan, PyMC and TensorFlow Probability. Its efficient handling of geometry via momentum is why it dominates modern Bayesian inference on continuous parameters.

8.8. Diagnostics

Verifying convergence of an MCMC run is a subtle empirical exercise. The standard tools are:

- **Trace plots** of $f(X_t)$ vs t : should look like white noise around a stationary mean.

- **Effective sample size (ESS):** $n_{\text{eff}} = n / (1 + 2 \sum_{k \geq 1} \hat{\rho}(k))$; ratios below 0.1 signal poor mixing.
- **Gelman–Rubin $\hat{R}$:** run M chains from over-dispersed starts and compare within-chain and between-chain variance. $\hat{R} < 1.01$ is the modern convergence threshold.

 Tip

Never trust an MCMC estimate from a single chain. Always run at least four in parallel, and inspect $\hat{R}$ before quoting a posterior number.

8.9. Exercises

Exercise 8.1

Simulate a Metropolis chain for a 2D banana-shaped target $\pi(x, y) \propto \exp(-x^2/10 - (y - x^2/10)^2/2)$. Estimate the effective sample size for the second component.

Solution

A random-walk Metropolis with isotropic proposal has $\text{ESS} \sim 10^2$ per 10^4 steps because the banana shape defeats isotropic scaling. HMC or preconditioning brings $\text{ESS} \sim 10^3$ or better. This is the standard illustration in the Stan documentation of why HMC dominates for non-elliptical posteriors.

Exercise 8.2

Prove that if $q(y | x) = q(x | y)$ then the Metropolis-Hastings acceptance probability reduces to $A = \min(1, \pi(y)/\pi(x))$, and verify detailed balance in this simplified form.

Solution

Substituting $q(y|x) = q(x|y)$ into the Hastings formula makes the q terms cancel. Detailed balance: $\pi(x)T(x, y) = \pi(x)q(y|x) \min(1, \pi(y)/\pi(x)) = q(y|x) \min(\pi(x), \pi(y)) = q(x|y) \min(\pi(x), \pi(y)) = \pi(y)T(y, x)$.

8.10. Callouts

 Note

Metropolis is *not* the fastest MCMC algorithm; it is the most general. For specific problems (Gaussian latent variables, hierarchical models), specialised samplers like NUTS or slice-sampling do orders of magnitude better.

 Warning

The MCMC estimator is *biased* by the initial burn-in period. Always discard a burn-in and check that discarding more doesn't change your estimate significantly.

 Caution

An MCMC chain that “looks converged” in a trace plot may still be missing modes of a multi-modal target. Trace plots detect *within-mode* stationarity, not *between-mode* mixing. Multi-modal posteriors need tempered chains or SMC.

 Important

Effective sample size is the *only* number that quantifies MCMC efficiency. Wall-clock time comparisons between samplers are only meaningful when normalised by ESS/sec.

8.11. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapters 8–9. MacKay, Chapter 29. Brooks, Gelman, Jones & Meng (eds.), *Handbook of Markov Chain Monte Carlo*, Chapters 1–5.

9. Gibbs sampling and high-dimensional inference

Metropolis is universal but blunt. When the target factorises well — Bayesian hierarchical models, mixture models, latent-variable models — a specialised algorithm called *Gibbs sampling* exploits the structure and dominates. This chapter introduces Gibbs sampling, illustrates it on mixture distributions, and works through a running high-dimensional inference example.

9.1. The Gibbs sampling algorithm

The idea: instead of proposing a full joint move and Metropolising, sample each block of variables *exactly* from its conditional given the others.

Definition 9.1 (Gibbs sampling). Given a joint density $\pi(\mathbf{x}) = \pi(x_1, \dots, x_d)$, the **Gibbs sampler** updates one coordinate at a time:

For $t = 0, 1, 2, \dots$:

1. Pick coordinate j (deterministically cycling or uniformly at random).
2. Sample $x_j^{(t+1)} \sim \pi(x_j \mid x_{-j}^{(t)})$ where x_{-j} denotes all coordinates except j .
3. Set $x_i^{(t+1)} = x_i^{(t)}$ for $i \neq j$.

Theorem 9.1 (Correctness of Gibbs sampling). *The Gibbs sampler leaves π invariant: if $\mathbf{X}^{(t)} \sim \pi$ then $\mathbf{X}^{(t+1)} \sim \pi$. Moreover under mild ergodicity assumptions the chain converges to π from any initial state.*

Proof. By construction the conditional draw preserves the joint: sampling $x_j \sim \pi(\cdot \mid x_{-j})$ and leaving x_{-j} fixed gives a joint that still integrates to $\pi(x_j, x_{-j}) = \pi(\mathbf{x})$. Detailed balance is verified coordinate-wise.

Remark. Gibbs sampling is a *special case* of Metropolis-Hastings with a proposal that is exactly the conditional distribution: acceptance is then always 1. This is the sense in which Gibbs “always accepts”.

9.2. When Gibbs sampling helps

Gibbs is efficient when the conditionals $\pi(x_j \mid x_{-j})$ have a familiar form (Gaussian, Dirichlet, categorical) so that we can sample from them exactly. Bayesian models with **conjugate priors** are the paradigm case.

Example 9.1 (Gibbs for Gaussian regression)

For the model $y_i \mid \beta, \sigma^2 \sim \mathcal{N}(\beta^\top \mathbf{x}_i, \sigma^2)$ with priors $\beta \sim \mathcal{N}(\mathbf{0}, \tau^2 I)$ and $\sigma^2 \sim \text{InvGamma}(\alpha, \beta)$:

- $\pi(\beta \mid \sigma^2, \mathbf{y}) = \mathcal{N}(\hat{\beta}, \Sigma_\beta)$ where $\Sigma_\beta = (\mathbf{X}^\top \mathbf{X} / \sigma^2 + I / \tau^2)^{-1}$, $\hat{\beta} = \Sigma_\beta \mathbf{X}^\top \mathbf{y} / \sigma^2$.
- $\pi(\sigma^2 \mid \beta, \mathbf{y}) = \text{InvGamma}(\alpha + n/2, \beta + \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2)$.

Alternating between the two conjugate updates samples the joint posterior in a few thousand iterations.

9.3. Mixture distributions

Latent-variable models are the sweet spot of Gibbs sampling. Consider a mixture of K Gaussians:

$$p(y_i \mid \mu, \sigma^2, \mathbf{w}) = \sum_{k=1}^K w_k \mathcal{N}(y_i; \mu_k, \sigma^2).$$

Example 9.2 (Gibbs for a Gaussian mixture)

Introduce a *latent* cluster indicator $z_i \in \{1, \dots, K\}$ for each observation with $p(z_i = k) = w_k$ and $p(y_i \mid z_i = k) = \mathcal{N}(\mu_k, \sigma^2)$. Gibbs sampling then alternates:

1. **Update** z_i : sample $z_i \sim \text{Categorical}(\gamma_{i,1}, \dots, \gamma_{i,K})$ with $\gamma_{i,k} \propto w_k \mathcal{N}(y_i; \mu_k, \sigma^2)$.
2. **Update** μ_k : sample μ_k from its Gaussian posterior given $\{y_i : z_i = k\}$.
3. **Update** $\mathbf{w}$: sample from the Dirichlet posterior given the cluster counts $\{|\{i : z_i = k\}|\}$.

The chain converges rapidly for well-separated components and mixes surprisingly slowly (multi-modality!) for overlapping ones.

9.4. A basic inference example

We now work through the running example from Prof. Allegra's notes: inferring a scalar signal θ from noisy binary measurements.

Definition 9.2 (Basic model). Observations $y_i \in \{0, 1\}$ for $i = 1, \dots, n$ are drawn as $y_i \sim \text{Bern}(\sigma(\theta))$ where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic function. Prior on θ : $\theta \sim \mathcal{N}(0, \tau^2)$.

The unnormalised posterior is

$$\pi(\theta \mid \mathbf{y}) \propto \exp(-\theta^2/(2\tau^2)) \prod_i \sigma(\theta)^{y_i} (1 - \sigma(\theta))^{1-y_i}.$$

This is not conjugate: no closed-form update. Two approaches:

Proposition 9.2 (Metropolis approach). Use a Gaussian random-walk proposal $\theta' = \theta + \sigma_p \xi$ and accept with the standard Metropolis ratio. Even with 10^5 samples the chain converges within a few hundred iterations for typical $n \sim 100$.

Proposition 9.3 (Analytic posterior expression). The posterior does not have a closed-form density but its log is

$$\log \pi(\theta \mid \mathbf{y}) = -\theta^2/(2\tau^2) + n_1 \theta - n \log(1 + e^\theta) + \text{const},$$

where $n_1 = \sum_i y_i$. The posterior is unimodal and concave, so a Laplace approximation around its mode gives a good analytic reference.

Example 9.3 (Simple numerical experiment)

Simulate $n = 200$ observations with true $\theta = 0.5$ and prior $\tau = 3$. The posterior is centred near 0.5 with standard deviation ~ 0.15 . Metropolis with $\sigma_p = 0.3$ yields an acceptance rate $\sim 40\%$ and an ESS/iteration of ~ 0.4 — respectable for a 1D problem.

9.5. Refined model: hierarchy

Push the example one level deeper. Suppose the true signal varies across G groups (a hierarchical model):

$$y_{gi} \sim \text{Bern}(\sigma(\theta_g)), \quad \theta_g \sim \mathcal{N}(\mu, \sigma_\theta^2), \quad \mu \sim \mathcal{N}(0, 10), \quad \sigma_\theta \sim \text{HalfNormal}(1).$$

Definition 9.3 (Refined hierarchical model). The joint posterior has $G+2$ parameters. Gibbs sampling exploits that θ_g are conditionally independent given the hyperparameters, so we can update them in parallel.

Proposition 9.4 (Metropolis-within-Gibbs). A hybrid scheme updates the hyperparameters (μ, σ_θ) from their conjugate Gaussian and half-Cauchy posteriors, and each θ_g via a small Metropolis step conditional on $\{y_{g1}, \dots\}$ and the current hyperparameters. This is the Metropolis-within-Gibbs algorithm, which dominates practical Bayesian inference before HMC took over in the 2010s.

Example 9.4 (Group-level shrinkage)

For small groups G with few observations each, hierarchical Bayes shrinks group estimates θ_g toward the global mean μ by exactly the same amount that James–Stein estimation does for a bank of Gaussians. This is one of the classical demonstrations that Bayes gives an operational meaning to shrinkage.

9.6. Modern developments

Since the 2010s the field has moved decisively toward Hamiltonian methods (HMC, NUTS) that scale better with dimension than either Metropolis or Gibbs. That said, Gibbs remains the algorithm of choice for models with discrete latent variables (mixtures, hidden Markov models, topic models like LDA), where Hamiltonian methods do not apply.

9.7. Exercises

Exercise 9.1

For the basic Bernoulli-logistic model above, derive the Laplace approximation to the posterior and compare its standard deviation with a Metropolis estimate on simulated data.

Solution

The mode $\hat{\theta}$ satisfies $-\hat{\theta}/\tau^2 + n_1 - n\sigma(\hat{\theta}) = 0$, solvable numerically. The Hessian at the mode is $-1/\tau^2 - n\sigma'(\hat{\theta}) = -1/\tau^2 - n\sigma(\hat{\theta})(1 - \sigma(\hat{\theta}))$. Laplace: $\pi(\theta | \mathbf{y}) \approx \mathcal{N}(\hat{\theta}, [-\text{Hess}]^{-1})$. Comparison to Metropolis with $n = 200$: agreement within 1% for the standard deviation.

Exercise 9.2

For a 2-component 1D Gaussian mixture with means at ± 3 and $\sigma^2 = 1$, show that Gibbs sampling on the latent indicators mixes well when the empirical bimodal structure is clear but fails to escape local modes when components strongly overlap. What is the physical explanation?

Solution

Non-mixing between local modes of Gibbs is a *label-switching* / *identifiability* issue: the Gibbs chain rarely proposes swaps between clusters when the current assignments are all consistent, so the chain gets stuck in one labelling. Fixing requires either parallel tempering, reversible-jump MCMC, or symmetric priors that identify clusters by ordering (e.g. $\mu_1 < \mu_2$).

9.8. Callouts**i Note**

Gibbs is *simple* and *exact* when conjugacy holds. When it doesn't, either derive an auxiliary-variable augmentation (Pólya–Gamma, slice) or fall back to Metropolis-within-Gibbs.

💡 Tip

Blocking correlated coordinates and updating them jointly (block Gibbs) dramatically improves mixing when strong correlations are present — the standard cure for slow mixing in hierarchical models.

⚠ Warning

Gibbs can catastrophically fail on multi-modal targets where individual conditionals are unimodal but the joint has multiple modes. Standard example: mixture models with symmetric priors.

! Important

The MCMC diagnostics of Chapter 7 (trace plots, ESS, $\hat{R}$) apply *without change* to Gibbs. Never trust a Gibbs chain without diagnostics — the fact that acceptance is 1 does not mean the chain has explored the target.

9.9. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapters 10–11. Gelman, Carlin, Stern, Dunson, Vehtari & Rubin, *Bayesian Data Analysis*, 3rd ed., CRC Press (2013), Chapters 11–12. Brooks et al. (eds.), *Handbook of MCMC*, Chapter 6.

10. A problem of high-dimensional inference

This chapter is a running example that ties Chapters 7–8 (MCMC, Gibbs) to a realistic Bayesian problem. It mirrors Prof. Allegra’s Chapter 11.

10.1. 9.1 Basic model

Suppose we observe n binary outcomes $y_i \in \{0, 1\}$ for $i = 1, \dots, n$, drawn as $y_i \sim \text{Bern}(\sigma(\theta))$ where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic function. The parameter θ has a Gaussian prior $\theta \sim \mathcal{N}(0, \tau^2)$.

10.1.1. Likelihood and Metropolis approach

The unnormalised posterior is

$$\pi(\theta \mid \mathbf{y}) \propto \exp(-\theta^2/(2\tau^2)) \prod_i \sigma(\theta)^{y_i} (1 - \sigma(\theta))^{1-y_i}.$$

There is no conjugate closed form. A Gaussian random-walk Metropolis proposal $\theta' = \theta + \sigma_p \xi$ with $\xi \sim \mathcal{N}(0, 1)$ and standard acceptance ratio gives a working sampler.

10.1.2. Analytic form of the posterior

Even without conjugacy, the log-posterior admits a closed expression:

$$\log \pi(\theta \mid \mathbf{y}) = -\frac{\theta^2}{2\tau^2} + n_1 \theta - n \log(1 + e^\theta) + \text{const},$$

with $n_1 = \sum_i y_i$. The posterior is unimodal and concave, so a **Laplace approximation** around its mode gives a good analytic reference for benchmarking MCMC results.

Proposition 10.1 (Laplace approximation). *Let $\hat{\theta} = \arg \max_\theta \log \pi(\theta \mid \mathbf{y})$ and $J = -\partial_\theta^2 \log \pi(\hat{\theta} \mid \mathbf{y}) > 0$. Then $\pi(\theta \mid \mathbf{y}) \approx \mathcal{N}(\hat{\theta}, J^{-1})$ to leading order in n . The mode $\hat{\theta}$ solves $-\hat{\theta}/\tau^2 + n_1 - n\sigma(\hat{\theta}) = 0$.*

10.1.3. Simple numerical example

Example 10.1

Simulate $n = 200$ observations with true $\theta = 0.5$ and prior $\tau = 3$. Metropolis with $\sigma_p = 0.3$ produces an acceptance rate $\sim 40\%$ and an effective sample size of ~ 4000 from 10^4 iterations. The posterior mean is close to 0.5 with standard deviation ~ 0.15 , agreeing with the Laplace approximation to 1%.

10.1.4. Gibbs sampling approach

For a scalar θ there is nothing to Gibbs-decompose; the algorithm reduces to Metropolis. Gibbs becomes useful in the refined model below.

10.2. 9.2 Refined model: hierarchical structure

Extend to G groups, each with its own signal θ_g :

$$y_{gi} \sim \text{Bern}(\sigma(\theta_g)), \quad \theta_g \sim \mathcal{N}(\mu, \sigma_\theta^2), \quad \mu \sim \mathcal{N}(0, 10), \quad \sigma_\theta \sim \text{HalfNormal}(1).$$

The joint posterior has $G + 2$ parameters.

10.2.1. Metropolis-within-Gibbs

Definition 10.1 (Metropolis-within-Gibbs). Alternate:

1. Update each θ_g via a small Metropolis step conditional on $\{y_{g1}, \dots\}$ and the current (μ, σ_θ) .
2. Update μ from its conjugate Gaussian posterior given $\{\theta_g\}$.
3. Update σ_θ from its (approximately conjugate) half-Cauchy posterior given $\{\theta_g - \mu\}$.

This scheme was the dominant sampler for hierarchical Bayesian models before HMC/NUTS became standard.

10.2.2. Gibbs sampling

If we replaced the Bernoulli likelihood by a Gaussian one (e.g. $y_{gi} \sim \mathcal{N}(\theta_g, \sigma_y^2)$), full conjugacy would hold and pure Gibbs (no Metropolis) would sample the entire posterior in closed form. This is the classical example of *hierarchical shrinkage* — an analytical version of James–Stein estimation.

Example 10.2 (Group-level shrinkage)

For small groups G with few observations each, hierarchical Bayes shrinks group estimates θ_g toward the global mean μ by exactly the same amount as James–Stein estimation. Bayes gives an operational meaning to shrinkage — via the hierarchical prior on θ_g .

10.3. Exercises

Exercise 10.1

Derive the Laplace approximation of the basic-model posterior. Explain why it is an *asymptotic* approximation whose error tends to zero as $n \rightarrow \infty$.

Solution

Compute $\hat{\theta}$ numerically as the root of $-\hat{\theta}/\tau^2 + n_1 - n\sigma(\hat{\theta}) = 0$; the second derivative is $-1/\tau^2 - n\sigma'(\hat{\theta})(1 - \sigma(\hat{\theta}))$. As $n \rightarrow \infty$ the posterior variance $J^{-1} \rightarrow 0$ so the distribution concentrates at $\hat{\theta}$; higher-order corrections to the Gaussian are $O(1/n)$, formalised in the Bernstein–von Mises theorem.

Exercise 10.2

Show that in the refined hierarchical model with Gaussian likelihood, the conditional distribution of θ_g given the data and hyperparameters is Gaussian. Give its mean and variance.

Solution

$\pi(\theta_g \mid \dots) \propto \exp(-\frac{1}{2\sigma_y^2} \sum_i (y_{gi} - \theta_g)^2 - \frac{1}{2\sigma_\theta^2} (\theta_g - \mu)^2)$. Completing the square gives

$$\theta_g \mid \dots \sim \mathcal{N}\left(\frac{n_g \bar{y}_g / \sigma_y^2 + \mu / \sigma_\theta^2}{n_g / \sigma_y^2 + 1 / \sigma_\theta^2}, (n_g / \sigma_y^2 + 1 / \sigma_\theta^2)^{-1}\right).$$

The posterior mean is a *precision-weighted average* of the group sample mean and the hyperparameter mean — exactly the shrinkage interpretation.

10.4. Callouts**i Note**

The high-dimensional-inference chapter is a testing ground for the algorithms of Chapters 7 and 8. Every method (Metropolis, Gibbs, Metropolis-within-Gibbs) should be benchmarked against the Laplace reference before believing its output.

💡 Tip

When designing a hierarchical model, always draw a *plate diagram* of the dependencies before writing code. Half of Bayesian bugs come from a mismatch between the intended model and its implementation.

⚠ Warning

Gibbs samplers for hierarchical models can mix very slowly when the variance hyperparameter σ_θ is close to zero (a “funnel” geometry). Reparameterisation (non-centred, i.e. $\theta_g = \mu + \sigma_\theta z_g$ with $z_g \sim \mathcal{N}(0, 1)$) is the standard fix.

10.5. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 11. Gelman, Carlin, Stern, Dunson, Vehtari & Rubin, *Bayesian Data Analysis*, 3rd ed., CRC Press (2013), Chapter 5.

11. Variational Bayes

MCMC is asymptotically exact but slow. **Variational Bayes** (VB) trades exactness for speed: it approximates a posterior by the closest member of a tractable family, converting inference into optimisation. This chapter mirrors Prof. Allegra's Chapter 12.

11.1. 10.1 The variational principle

Given a target posterior $p(\mathbf{z} \mid \mathbf{x})$ we cannot compute directly, choose a tractable family $\mathcal{Q} = \{q_\phi(\mathbf{z})\}$ parameterised by ϕ , and find the member closest to p in KL divergence:

$$\phi^* = \arg \min_{\phi} D(q_\phi(\mathbf{z}) \parallel p(\mathbf{z} \mid \mathbf{x})).$$

Theorem 11.1 (Evidence lower bound (ELBO)). *The KL divergence $D(q \parallel p)$ equals $\log p(\mathbf{x}) - \mathcal{L}(\phi)$ where*

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}[\log p(\mathbf{x}, \mathbf{z})] - \mathbb{E}_{q_\phi}[\log q_\phi(\mathbf{z})]$$

*is the **evidence lower bound**. Since $\log p(\mathbf{x})$ does not depend on ϕ , maximising $\mathcal{L}$ is equivalent to minimising the KL.*

The ELBO decomposes as $\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{x} \mid \mathbf{z})] - D(q_\phi(\mathbf{z}) \parallel p(\mathbf{z}))$ — a reconstruction fit minus a complexity penalty.

11.2. 10.2 Mean-field approximation

The simplest tractable family assumes independence across coordinates:

Definition 11.1 (Mean-field variational family). The **mean-field family** on $\mathbb{R}^d$ is

$$q(\mathbf{z}) = \prod_{j=1}^d q_j(z_j).$$

Under mean field, the optimal factor for coordinate j has an explicit form.

Proposition 11.2 (Mean-field CAVI update). *The mean-field ELBO is maximised by iterating*

$$q_j^*(z_j) \propto \exp(\mathbb{E}_{q_{-j}}[\log p(\mathbf{x}, \mathbf{z})]),$$

*i.e. each factor's log is the expected joint log-density under all other factors. This is **coordinate ascent variational inference (CAVI)** — the analogue of Gibbs sampling but with expectations replacing samples.*

11.3. 10.3 Example: one Gaussian

Example 11.1

Model: $x_i \sim \mathcal{N}(\mu, 1/\lambda)$ with priors $\mu \sim \mathcal{N}(0, 1/\lambda_0)$, $\lambda \sim \text{Gamma}(a_0, b_0)$. Mean-field factorisation $q(\mu, \lambda) = q(\mu)q(\lambda)$:

- $q^*(\mu) = \mathcal{N}(m_n, 1/(\lambda_n + \lambda_0))$ with $\lambda_n = \mathbb{E}[\lambda] \cdot n$ and m_n a weighted mean.
- $q^*(\lambda) = \text{Gamma}(a_n, b_n)$ with $a_n = a_0 + n/2$ and b_n a function of $\mathbb{E}[(x_i - \mu)^2]$.

CAVI alternates between the two updates. Converges in ~ 10 iterations to a solution that closely matches the exact NIG posterior.

11.4. 10.4 Example: mixture distributions

For a K -component mixture model, mean-field VB introduces a variational cluster assignment $q(z_i = k)$ and Gaussian variational distributions on each mean μ_k . The result is the **variational EM algorithm** — a Bayesian extension of standard EM that avoids over-fitting.

11.5. 10.5 Variational autoencoders (VAE)

The most influential modern application of variational Bayes is the **variational autoencoder** (Kingma & Welling, 2014):

Definition 11.2 (Variational autoencoder). A **VAE** models data $\mathbf{x}$ as generated by a latent code $\mathbf{z}$ through a decoder network $p_\theta(\mathbf{x} | \mathbf{z})$ with $\mathbf{z} \sim \mathcal{N}(0, I)$. The variational posterior is parameterised by an encoder network $q_\phi(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\mu_\phi(\mathbf{x}), \Sigma_\phi(\mathbf{x}))$. Training jointly optimises the ELBO w.r.t. (ϕ, θ) .

The reparameterisation trick $\mathbf{z} = \mu_\phi + \Sigma_\phi^{1/2} \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$ makes the ELBO differentiable through the sampling step, so stochastic gradients propagate cleanly through both networks.

Remark. VAEs revolutionised generative modelling: for the first time a single differentiable objective produced samples of natural images, speech, and molecules. Their descendants (-VAE, VQ-VAE, hierarchical VAE) remain competitive with diffusion models for certain modalities.

11.6. Exercises

Exercise 11.1

Derive the ELBO decomposition $\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{x} | \mathbf{z})] - D(q\|p(\mathbf{z}))$.

Solution

$$\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{x}, \mathbf{z}) - \log q(\mathbf{z})] = \mathbb{E}_q[\log p(\mathbf{x} | \mathbf{z}) + \log p(\mathbf{z}) - \log q(\mathbf{z})] = \mathbb{E}_q[\log p(\mathbf{x} | \mathbf{z})] - D(q\|p(\mathbf{z})).$$

Exercise 11.2

Show that mean-field VB *underestimates* posterior variance for correlated targets. Give a two-dimensional Gaussian example.

Solution

For a Gaussian target with covariance Σ and mean-field family $q = \mathcal{N}(\mu, \text{diag}(\sigma^2))$, the optimal variance of q_j is $1/(\Sigma^{-1})_{jj}$, which is *less* than Σ_{jj} whenever off-diagonal elements are non-zero. The intuition: mean field cannot capture correlations, so it compensates by inflating the “certainty” of each coordinate.

11.7. Callouts

i Note

Variational Bayes is *biased* by construction: the mean-field family cannot represent the true posterior in general. For applications where posterior *shape* matters (uncertainty quantification, decision theory), use HMC; for applications where posterior *mean* is enough (predictions, model comparison), VB is fine.

💡 Tip

When implementing CAVI, initialise the variational parameters using a rough clustering of the data. Random initialisation often gets stuck in local ELBO maxima.

⚠ Warning

The mean-field variance under-estimation is a serious limitation for tasks like Bayesian neural networks where calibrated uncertainty is the whole point. Use *normalising flows* or *low-rank* variational families to escape it.

11.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 12. Blei, Kucukelbir & McAuliffe, *Variational Inference: a review for statisticians*, J. Am. Stat. Assoc. **112**, 859 (2017). Kingma & Welling, *Auto-encoding variational Bayes*, ICLR (2014).

12. Hamiltonian Monte Carlo

Random-walk Metropolis (Chapter 7) mixes slowly in high dimensions. Hamiltonian Monte Carlo (HMC) uses ideas from classical mechanics to propose distant, high-acceptance moves. This chapter mirrors Prof. Allegra's Chapter 13.

12.1. 11.1 Hamiltonian dynamics

Augment the state $\mathbf{q} \in \mathbb{R}^d$ with an auxiliary momentum $\mathbf{p} \in \mathbb{R}^d$, and define the **Hamiltonian**

$$H(\mathbf{q}, \mathbf{p}) = U(\mathbf{q}) + K(\mathbf{p}) = -\log \pi(\mathbf{q}) + \frac{1}{2} \mathbf{p}^\top M^{-1} \mathbf{p},$$

where π is the target distribution, $U = -\log \pi$ is the *potential energy*, and K is a Gaussian *kinetic energy* with mass matrix M .

Definition 12.1 (Hamilton's equations). The Hamiltonian dynamics

$$\dot{\mathbf{q}} = \partial_{\mathbf{p}} H = M^{-1} \mathbf{p}, \quad \dot{\mathbf{p}} = -\partial_{\mathbf{q}} H = -\nabla U(\mathbf{q})$$

preserve the joint density $e^{-H(\mathbf{q}, \mathbf{p})}$ (Liouville's theorem) and conserve energy ($dH/dt = 0$).

12.2. 11.2 Numerical integration: leapfrog

The continuous Hamiltonian flow cannot be integrated exactly. The **leapfrog** scheme is a symplectic integrator that preserves the *volume* of phase space and conserves H up to $O(\Delta t^2)$ error:

Definition 12.2 (Leapfrog step). Given step size ε :

1. $\mathbf{p} \leftarrow \mathbf{p} - \frac{\varepsilon}{2} \nabla U(\mathbf{q})$ (half-step in momentum).
2. $\mathbf{q} \leftarrow \mathbf{q} + \varepsilon M^{-1} \mathbf{p}$ (full step in position).
3. $\mathbf{p} \leftarrow \mathbf{p} - \frac{\varepsilon}{2} \nabla U(\mathbf{q})$ (half-step in momentum).

Composition of L leapfrog steps gives a proposal $(\mathbf{q}, \mathbf{p}) \rightarrow (\mathbf{q}', \mathbf{p}')$ with symplectic Jacobian equal to 1.

12.3. 11.3 The HMC algorithm

Definition 12.3 (Hamiltonian Monte Carlo). Given current $\mathbf{q}$:

1. Sample momentum $\mathbf{p} \sim \mathcal{N}(0, M)$.
2. Run L leapfrog steps of size ε starting from $(\mathbf{q}, \mathbf{p})$, giving $(\mathbf{q}', \mathbf{p}')$.

3. Accept the proposal $\mathbf{q}'$ with probability

$$A = \min(1, \exp(H(\mathbf{q}, \mathbf{p}) - H(\mathbf{q}', \mathbf{p}'))).$$

Otherwise return $\mathbf{q}$.

The acceptance probability depends only on the (small) integration error of the leapfrog scheme, not on the distance travelled. This is why HMC can propose moves of $O(1)$ magnitude in high dimension while keeping acceptance ~ 0.7 .

Theorem 12.1 (HMC correctness). *Detailed balance holds: the HMC kernel leaves the joint distribution $\pi(\mathbf{q}) \cdot \mathcal{N}(\mathbf{p}; 0, M)$ invariant. Consequently the marginal on $\mathbf{q}$ is π .*

12.4. 11.4 Advantages of HMC

The essential reason HMC dominates in modern Bayesian inference is its dimension scaling:

Proposition 12.2 (HMC scaling). *For a target with sufficient regularity, the optimal number of leapfrog steps scales as $L \sim d^{1/4}$ and the optimal step size as $\varepsilon \sim d^{-1/4}$, giving an effective sample size per gradient evaluation of $O(d^{-1/4})$. Random-walk Metropolis, by contrast, has $ESS/eval \sim O(d^{-1})$.*

The one-order-of-magnitude improvement in scaling is what made HMC the default sampler for continuous parameters.

Remark (NUTS). The No-U-Turn Sampler (Hoffman & Gelman, 2014) automates HMC by dynamically choosing L so the trajectory stops just before it starts curving back. NUTS is the algorithm underlying Stan, PyMC, and TensorFlow Probability.

12.5. 11.5 Practical considerations

12.5.1. Mass matrix

The mass matrix M acts as a preconditioner. For Gaussian-like targets, $M = \text{diag}(\text{Var}(q_j)^{-1})$ or $M = \Sigma_\pi^{-1}$ (the *inverse target covariance*) makes the Hamiltonian isotropic and improves mixing dramatically. Modern implementations estimate M from an initial warm-up phase.

12.5.2. Step size adaptation

If ε is too large the leapfrog error blows up and acceptance goes to zero; too small and L blows up. Dual averaging adapts ε during warm-up to hit a target acceptance ~ 0.8 .

12.5.3. Non-continuous targets

HMC requires a differentiable log-density. For discrete parameters or heavy-tailed distributions, HMC fails; Gibbs sampling or ensemble MCMC methods remain the tools of choice.

12.6. Exercises

Exercise 12.1

For a Gaussian target $\pi(\mathbf{q}) = \mathcal{N}(0, \Sigma)$ show that Hamiltonian dynamics with $M = \Sigma^{-1}$ decouples into independent harmonic oscillators along the eigenvectors of Σ .

Solution

The Hamiltonian is $H = \frac{1}{2}\mathbf{q}^\top \Sigma^{-1} \mathbf{q} + \frac{1}{2}\mathbf{p}^\top \Sigma \mathbf{p}$. Diagonalise $\Sigma^{-1} = O\Lambda O^\top$ and transform $\tilde{\mathbf{q}} = O^\top \mathbf{q}$, $\tilde{\mathbf{p}} = O^\top \mathbf{p}$. Then $H = \frac{1}{2}\tilde{\mathbf{q}}^\top \Lambda \tilde{\mathbf{q}} + \frac{1}{2}\tilde{\mathbf{p}}^\top \Lambda^{-1} \tilde{\mathbf{p}}$, a sum of independent harmonic oscillators. Each mode is analytically integrable — HMC is exact in this special case.

Exercise 12.2

Show that the leapfrog integrator is reversible: swapping the sign of momentum and integrating backwards returns to the starting point.

Solution

Direct verification of the three-step definition: applying it with $\varepsilon \rightarrow -\varepsilon$ inverts each step, and by symmetry of the composition the whole trajectory reverses exactly.

12.7. Callouts

Note

HMC is *the* modern MCMC sampler for continuous parameters. If your model is smooth and differentiable, use HMC/NUTS; use random-walk Metropolis only for discrete parameters or as a baseline.

Tip

When HMC has trouble mixing (low ESS, divergent transitions), reparameterise the model to make its geometry more isotropic. Non-centred parameterisations of hierarchical models are the canonical example.

Important

The energy conservation of leapfrog holds *up to* $O(\varepsilon^2)$ per step but *without drift*: energy oscillates but does not systematically increase. This is what makes long trajectories possible without a Metropolis correction blowing up.

Warning

HMC requires evaluating $\nabla \log \pi(\mathbf{q})$ at every leapfrog step. If your log-density is expensive (e.g. an ODE solve inside the likelihood), HMC can be prohibitively slow.

12.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 13. Neal, *MCMC using Hamiltonian dynamics*, Ch. 5 of Brooks et al. (eds.), *Handbook of MCMC*, CRC (2011). Betancourt, *A conceptual introduction to Hamiltonian Monte Carlo*, arXiv:1701.02434.

13. Approximate Bayesian computation

MCMC and VB all require the *likelihood* $p(\mathbf{y} \mid \theta)$ to be evaluable. What if the likelihood is only *simulable* — i.e. we can generate data from the model but cannot compute its density? That is the world of **approximate Bayesian computation** (ABC). This chapter mirrors Prof. Allegra’s Chapter 14.

13.1. 12.1 Motivation: intractable likelihoods

Many scientific models are defined by a simulator: a stochastic dynamical system that emits synthetic data given parameters. The associated likelihood may be:

- an intractable marginal (e.g. a hidden-state model where the joint is tractable but the observed marginal is not),
- an unnormalised density with an intractable partition function (e.g. certain physics-inspired models),
- entirely implicit (e.g. agent-based simulations in ecology and epidemiology).

For such models MCMC’s Metropolis acceptance ratio $\pi(\theta')/\pi(\theta)$ cannot be computed. ABC solves this by working *directly* with simulations.

13.2. 12.2 ABC rejection sampling

Definition 13.1 (ABC rejection algorithm). Given observed data $\mathbf{y}_{obs}$, prior $\pi(\theta)$, simulator $S(\theta)$, distance $d(\cdot, \cdot)$, and tolerance ϵ :

Repeat until N samples are accepted:

1. Draw $\theta^{(i)} \sim \pi(\theta)$.
2. Simulate $\mathbf{y}^{(i)} = S(\theta^{(i)})$.
3. If $d(\mathbf{y}^{(i)}, \mathbf{y}_{obs}) < \epsilon$, accept $\theta^{(i)}$.

The accepted samples are distributed as $\pi_\epsilon(\theta \mid \mathbf{y}_{obs})$, which converges to $\pi(\theta \mid \mathbf{y}_{obs})$ as $\epsilon \rightarrow 0$.

Remark. ABC replaces the (unavailable) likelihood ratio by an *indicator* on distance — a very general trick. Its downside is that acceptance is $\ll 1$ when ϵ is small, so ABC-rejection is impractical for anything but toy problems.

13.3. 12.3 ABC-MCMC and ABC-SMC

Two improvements make ABC practical:

Definition 13.2 (ABC-MCMC). Combine ABC with a Metropolis proposal on θ : propose $\theta' \sim q(\theta' \mid \theta)$,

simulate $\mathbf{y}'$, and accept with probability

$$A = \min\left(1, \frac{\pi(\theta') q(\theta | \theta')}{\pi(\theta) q(\theta' | \theta)}\right) \cdot \mathbf{1}[d(\mathbf{y}', \mathbf{y}_{obs}) < \epsilon].$$

The chain concentrates on regions of the parameter space where simulations resemble the data.

Definition 13.3 (ABC-SMC). Sequential Monte Carlo variant: start with ϵ_1 large and gradually anneal $\epsilon_t \searrow \epsilon_{final}$, reweighting particles as ϵ shrinks. Widely used in population genetics and epidemiology.

13.4. 12.4 Summary statistics

For high-dimensional data $\mathbf{y}$, the raw distance $d(\mathbf{y}', \mathbf{y}_{obs})$ is nearly always huge; ABC becomes uninformative. The standard remedy is to summarise $\mathbf{y}$ by a low-dimensional statistic $T(\mathbf{y})$ and require $d(T(\mathbf{y}'), T(\mathbf{y}_{obs})) < \epsilon$.

Proposition 13.1 (Sufficient statistics). *If T is a sufficient statistic for θ , ABC on T has the same posterior as ABC on the raw data, i.e. no information is lost. In practice sufficient statistics rarely exist; carefully-chosen summary statistics (empirical moments, quantiles, autocorrelations) are used instead.*

Remark (Learned summary statistics). Modern ABC uses neural networks to learn near-sufficient summary statistics automatically. **Neural posterior estimation** (Papamakarios & Murray, 2016) trains a conditional density $q_\phi(\theta | T(\mathbf{y}))$ to match the true posterior, blending ABC with variational Bayes.

13.5. 12.5 Applications

Example 13.1 (Population genetics)

The coalescent model produces gene genealogies whose likelihood is intractable but easily simulable. ABC is the standard tool for inferring demographic parameters (population size, migration rates, split times) from DNA data. Beaumont et al. (2002) is the seminal reference.

Example 13.2 (Epidemic dynamics)

Agent-based models of infectious-disease spread are intractable but simulable. ABC-SMC was used during COVID-19 to fit parameters of the UK's SPI-M models given hospitalisation time series.

13.6. Exercises

Exercise 13.1

Show that ABC rejection with $\epsilon = 0$ (and continuous data) has acceptance probability zero.

Solution

For continuous $\mathbf{y}$, $P(\mathbf{y}' = \mathbf{y}_{obs}) = 0$. The Dirac-delta indicator is never triggered. Hence any positive tolerance $\epsilon > 0$ is a *necessary* approximation for continuous data.

Exercise 13.2

For a Gaussian likelihood $\mathbf{y} \mid \theta \sim \mathcal{N}(\theta, I)$ with a Gaussian prior, show that ABC with sufficient statistic $T = \bar{y}$ recovers the exact posterior in the limit $\epsilon \rightarrow 0$.

Solution

$\bar{y}$ is a sufficient statistic for θ . As $\epsilon \rightarrow 0$ the ABC posterior converges to the true posterior conditional on $\bar{y} = \bar{y}_{obs}$, which by sufficiency equals the true posterior conditional on the full data.

13.7. Callouts

Note

ABC is *the* tool for likelihood-free inference. Every scientific problem where a simulator is the model's definition (not a computational shortcut) is a candidate.

Tip

When designing an ABC study, invest time in choosing summary statistics. The success of ABC hinges on their sufficiency; a bad choice makes the posterior arbitrarily far from the truth.

Warning

ABC posteriors are *biased* estimators of the true posterior. Studies that report ABC results should always report the tolerance ϵ and, ideally, a robustness check as ϵ is decreased.

Caution

Simulation-based inference (SBI), including modern neural variants, has largely superseded classical ABC for high-dimensional problems. Consult the `sbi` Python library for state-of-the-art tools.

13.8. Sources

Prof. Allegra, *Notes on Information Theory and Inference*, Chapter 14. Beaumont, Zhang & Balding, *Approximate Bayesian computation in population genetics*, *Genetics* **162**, 2025 (2002). Cranmer, Brehmer & Louppe, *The frontier of simulation-based inference*, *PNAS* **117**, 30055 (2020).

